A single nested sequence of finite Markovization times approximates the Markov-quantile process

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Editorial note. This task counts Theorem B in the author stated finite-dimensional convergence sense. No arbitrary dense time set, full path topology, continuous version or strong Markov assertion is claimed. The detailed theorem excerpt stops at(a),(b); its(c) classification is not assumed.

Editorial note. Increasing kernel is used in Proposition3.11(b), printed here. The unused probability-valued monotone extension to every point outside the initial support in3.11(c) is not asserted. All actual level kernels act on[0,1]; general quantile images use almost-sure versions.

Editorial note. The source bounded-density closure wording1095-1104 is used only with a common bound. In1345 a fixed input theta<=M*mu has theta P_n<=M*nu because the target marginal is fixed; thus the required uniform bound is present. No closure claim for arbitrarily unbounded densities is assumed.

Editorial note. At970 the general d-dimensional rectangle increment has the usual sign(-1)^(d-|epsilon|); the printed even-parity convention is correct in dimension2, the dimension of the directed level-coupling construction. At993 the mixed-increment condition must hold for the infimum CDF, as991 and the actual proof1017 explain, not merely for each individual CDF.

Editorial note. At1605 the CDF comparison is reversed in print: the stochastic monotonicity proved1582 gives F[L_R]<=F[L_R-prime] when R-prime subset R, consistent with the stated lower-orthant order1591. Original formulas are retained.

Editorial note. At1817 the finite approximating choices can and must retain the preceding finite set when moving to accuracy n, as required by1811. Equivalently take cumulative finite unions of the displayed choices; monotonicity1804/1591 preserves every approximation. This makes the author intended nested diagonal construction explicit.1876-1877 uses an undefined R-prime_n; the directly preceding1811 approximation and1804 limit already identify the limit for the chosen R_n. Do not infer a common time set from pairwise existence alone.

Editorial note. At1832 the second factor concerns(t,s-prime), as correctly written in the next line1833, not the repeated(s,t) interval. The contradiction remains the author composition/continuity argument.

Editorial note. In the atomic covariance proof,1915 suppresses the Lebesgue factor in the constant-density measure, and1919 reverses its displayed indicator inequality. Both are fixed by1915-1917 and the nonnegative covariance identity. At1957 both lower-orthant kernel test functions are decreasing, not increasing; the same covariance inequality holds for two functions with the same direction, consistently with2062-2063.

Editorial note. The multivariate targets in1925/2054 have the stated coordinate marginals but are not assumed independent. The actual kernel integrals1941-1966 and the chain formula2063 use their full joint laws. The printed tensor-product target notation cannot be read as an extra product-law hypothesis.

Editorial note. In1994-2016, m/n both refer to the number of Markovization times; all blocks B_0,...,B_m are included. The explicit finite endpoint products in2000-2002 determine the block law; the L_[r_k,r_(k+1)] label there is not a replacement by all intervening real times or an identification of two- and three-coordinate laws. At2005 the defining q_r has starting measure lambda, as1454-1456 specifies, not mu_r.2016 ends its interior list atell_k, as correctly written2021, and2051 ends the chain withs_(k-1),s_k.

Editorial note. Original references to the simultaneous two-point/whole-process equality are not treated as proof shortcuts: the full finite-level pushforward argument1990-2036 and the atomic-factorization argument2050-2064 are retained.

Editorial note. Source1375 says the sequence is increasing: the needed and definitionally supplied fact is that each function tilde(h)_n is increasing in its spatial variable, not that n maps to that function is increasing.

Editorial note. The pointwise proof1373-1385 uses [0,1] notation while the concatenation result is on R. Its bounded-coordinate argument transports through any fixed increasing homeomorphism R to(0,1); this ordinary coordinate convention and the already printed quantile transform handle general probability marginals. Independent review must confirm that no substantive missing author argument is concealed by this convention.

Editorial note. At1381 the symmetric lower-limit inequality is printed <=; the symmetric claim needed for convergence is >=, uniquely fixed by the limsup bound being proved and the target conclusion. At1379 the displayed difference of half-line indicators gives[a,b), not[a,b]; either endpoint convention is allowed among monotone test functions, including1_(b,infinity) for the closed version. The originals remain visible.

Editorial note. At1389 the displayed order P-prime circ P reverses the typed concatenation. The defining marginals and actual proof1395,1399 unambiguously use P circ P-prime. At1402 tilde(h)_n means P-prime_n dot h, as the preceding integral1399 explicitly shows. At1435 the repeated index i+i and last n are local index slips determined by the adjacent projection i,i+1 and chain length d. No author formula is silently overwritten.

Editorial note. At1807 the final displayed distance has the finite approximant as its first entry. The initial monotone-metric remark in the same proof directly bounds rho(L_Rtau,L_R) by rho(L_Rprime,L_R); this is the quantity required for convergence. The source order sandwich and proof are retained.

Editorial scope. One Theorem B counting unit. The detailed Theorem4.21(a),(b) is supporting proof; its separate(c) classification/necessity statements and later qualitative examples are not additional tasks. The full source1451-1705 construction is retained to define the claimed limit without assuming the sought approximation.

Allowed standard prerequisites (editorial).

- Disintegration, Fubini, dominated convergence, monotone classes and Kolmogorov-Daniell extension. General probability foundations; the particular consistency of level kernels and preservation of the Markov property after quantile pushforward remain proved.
- Portmanteau theorem, CDF characterization, Lebesgue differentiation and elementary monotone-function facts. The source proves the stronger fixed-marginal uniform CDF metric and finite-concatenation continuity; neither is supplied as a free research input.
- Quantile transform, stochastic order and increasing coordinate homeomorphisms R to(0,1). Interpret the author bounded-coordinate continuity proof for general real marginals. No common-time approximation theorem or atomic distributivity theorem is supplied as background.

Source reference. Original Theorem A: sources/markov-quantile-jep-2022-177/source0.tex, source lines 295–325. Historical existence/minimality theorem; the exact level-kernel construction required for Theorem B is proved in the retained1451-1705 source. No missing global approximation is supplied by this pointer.

Source reference. Original section A(a)(iv): sources/markov-quantile-jep-2022-177/source0.tex, source lines 313–317. Individual pairwise limiting description recalled only as context. The retained construction1644-1705 states and proves that description; the new common-time and full-process obligation is proved separately.

Source reference. Original section 4.21(c): sources/markov-quantile-jep-2022-177/source0.tex, source lines 1777–1786. Unselected later classification of acceptable time sets. The counted Theorem B uses only the included(a),(b), whose entire proof is retained.

Source reference. Original Proposition 3.11: sources/markov-quantile-jep-2022-177/source0.tex, source lines 1055–1086. The required almost-sure predicate(b) is explicitly printed1062-1063; its implication to order preservation is printed1079. The unused global extension(c) is not assumed.

Source reference. Original section 3.11(a): sources/markov-quantile-jep-2022-177/source0.tex, source lines 1058–1059. Order-preservation reading is exactly the consequence proved in the retained1079; this pointer introduces no free local theorem.

Source reference. Original section 3.11(b): sources/markov-quantile-jep-2022-177/source0.tex, source lines 1061–1063. The complete predicate is directly included1062-1063; only its label lies outside the selected plain-text definition.

Source reference. Original section 4.8: sources/markov-quantile-jep-2022-177/source0.tex, source lines 1545–1563. Author explicitly identifies the remark and figures as

qualitative and skippable1543. The actual level-kernel monotonicity proof1521-1533 and1578-1583 is retained.

Source reference. Original section 3.9: sources/markov-quantile-jep-2022-177/source0.tex, source lines 1034–1050. General stochastic-supremum explanation; the CDF definition/order and directed existence lemma with complete proof are directly supplied934-1031. No extra supremum step is hidden here.

Source reference. Original section 3.1: sources/markov-quantile-jep-2022-177/source0.tex, source lines 931–1031. Section pointer only; actual order definitions and required supremum proof are included.

Scope of the simultaneous approximation theorem. Let $(\mu_t)_{t\in\mathbb{R}}$ be an arbitrary family of probability measures on $\mathbb{R}$. The quantile law is obtained from one uniform variable by the coordinate quantile maps. The Markov–quantile law is the Markov law whose pair marginals are constructed below from the level kernels $L_{[s,t]}$. Convergence in the counted Theorem B means convergence of every finite-dimensional marginal, as specified in the author's Reminder following that theorem. Increasing kernels are used in the author's almost-sure conditional-expectation sense, not as a demand for a probability-valued monotone extension outside the support of the starting marginal. All finite time tuples are ordered increasingly. The supporting excerpt of the detailed theorem includes parts (a) and (b) only; its further classification of admissible time sets in part (c) is not counted.

Notation 0.1. For all measurable space E, $\mathcal{M}(E)$ and $\mathcal{P}(E)$ are the spaces of measures and probability measures on E. If $\mathcal{T}' \subset \mathcal{T}$, $\mathrm{proj}^{\mathcal{T}'}$ is the projection $\prod_{\tau\in\mathcal{T}} E_\tau \to \prod_{\tau\in\mathcal{T}'} E_\tau$; in case $\mathcal{T} = \{\tau_1, \dots, \tau_m\}$ is finite, $\mathrm{proj}^{\tau_1,\dots,\tau_m}$ means $\mathrm{proj}^{\{\tau_1,\dots,\tau_m\}}$. When $P \in \mathcal{P}\big(\prod_{\tau\in\mathcal{T}} E_\tau\big)$ and $s < t$, P^s stands for $(\mathrm{proj}^s)_\# P$ and $P^{s,t}$ for $(\mathrm{proj}^{s,t})_\# P$, and $\mathrm{Marg}((\mu_\tau)_{\tau\in\mathcal{T}})$ denotes $\{P \in \mathcal{P}\big(\prod_{\tau\in\mathcal{T}} E_\tau\big) : \forall\, \tau \in \mathcal{T}, (\mathrm{proj}^\tau)_\# P = \mu_\tau\}$. When not otherwise specified, what we call the marginals of P are its marginals P^s on a single factor.

Reminder 0.2. A sequence $(P_n)_n$ of (probability) measures on some measurable topological space E *converges weakly* to P if, for all bounded continuous function f, $\int f \mathrm{d}P_n \to \int f \mathrm{d}P$. For $E = \mathbb{R}^{\mathbb{R}}$ with the weak topology, this convergence amounts to the weak convergence of all finite marginals.

Everywhere E, E', E'' etc. are topological spaces (or sometimes Polish spaces) and $\mathcal{B}(E)$, $\mathcal{B}(E')$ and $\mathcal{B}(E'')$ their Borel σ-algebras.

Definition 0.3. A probability kernel, or kernel k from E to E' is a map

$$k : E \times \mathcal{B}(E') \to [0, 1]$$

such that $k(x, \cdot)$ is a probability measure on E' for every x in E and $k(\cdot, B)$ is a measurable map for every $B \in \mathcal{B}(E')$.

Probability kernels are usually interpreted as transition matrices, see Remark 0.5: after one step a particle at x in E arrives at a random position in E', distributed with respect to $k(x, \cdot)$. We often have that interpretation in mind.

Remark/Notation 0.4. Every transport plan $P \in \mathcal{P}(E \times E')$ can be disintegrated with respect to its first marginal $P^1 := (\mathrm{proj}^1)_\# P$ and a kernel that we denote by k_P, defined from E to E', so that:

$$\iint f(x,y)\,\mathrm{d}P(x,y) = \int \left(\int f(x,y)\,k_P(x,\mathrm{d}y) \right) \mathrm{d}P^1(x)$$

for every bounded continuous function f. Observe that $x \mapsto k(x,\cdot)$ is P^1-almost surely uniquely determined.

0.0.1 Composition and action of kernels

Kernels k from E to E' and k' from E' to E'' can be composed as follows:

$$(k \,.\, k')(x,A) = \int_{E'} k'(y,A) k(x,\mathrm{d}y).$$

Similarly, acting on the right, kernels from E to E' *transport*, or *send* positive measures θ on E on positive measures on E'. Acting on the left, they send (adequately integrable) functions $f : E' \to \mathbb{R}$, on functions $E \to \mathbb{R}$:

$$(\theta \,.\, k)(A) = \int_{E'} k(y,A)\theta(\mathrm{d}y) \quad \text{and:} \quad (k \,.\, f)(x) = \int_{E'} f(y) k(x,\mathrm{d}y).$$

Associativity holds, e.g., $(k \,.\, k') \,.\, k'' = k \,.\, (k' \,.\, k'')$, and $\theta \,.\, (k \,.\, f) = (\theta \,.\, k) \,.\, f$, where the action of measures on functions is the obvious one. This is consistent with the following remark.

Remark 0.5. We recall the usual interpretation of the composition as matrix multiplication. If $E = \{x_i\}_{i=1}^n$, $E' = \{y_j\}_{j=1}^{n'}$, $E'' = \{z_k\}_{k=1}^{n''}$ are finite, a measure $\theta \in \mathcal{M}(E)$ is a row vector $(\theta(\{x_i\}))_{i=1}^n = (\theta_i)_{i=1}^n$, a kernel k from E to E' is a matrix $k = ((k_{i,j})_{i=1}^n)_{j=1}^{n'}$, where $(k_{i,j})_{j=1}^{n'}$ is the measure $k(x_i,\,\cdot\,) \in \mathcal{M}(E')$, viewed as a vector, and a function f from E'' to $\mathbb{R}$ is a (column) vector $f = (f(z_j))_{j=1}^{n''}$. Then, taking $k' = ((k'_{j,k})_{j=1}^{n'})_{k=1}^{n''}$ a kernel from E' to E'', $\theta \,.\, k$, $k \,.\, k'$ and $k' \,.\, f$ introduced above have the same sense as products of matrices.

Notation 0.6. We denote by id_E the identity kernel (that acts trivially) $(x,B) \mapsto \delta_x(B) = \mathbb{1}_B(x)$.

Notation 0.7. With $\mu \in \mathcal{M}(E)$ and k a kernel from E to E' is naturally associated the law $\mathrm{Joint}(\mu,k) \in \mathcal{M}(E \times E')$, having μ as first marginal and the family $(k(x_0,\cdot))_{x_0 \in E}$ as laws (on E') conditioned by $x_0 \in E$:

$$\forall\, B, B' \in \mathcal{B}(E) \times \mathcal{B}(E'), \quad \mathrm{Joint}(\mu,k)(B \times B') = \int_B k(x,B')\mathrm{d}\mu(x).$$

In particular, $P = \mathrm{Joint}(P^1, k_P)$.

0.0.2 Composition of transport plans

If $P \in \mathrm{Marg}(\mu,\mu')$ and $Q \in \mathrm{Marg}(\mu',\mu'')$, we can compose them in a similar way as we compose kernels, getting the product:

$$P \,.\, Q := \mathrm{Joint}(\mu, k_P \,.\, k_Q) \in \mathrm{Marg}(\mu,\mu''), \quad \text{so that: } k_{P.Q} = k_P \,.\, k_Q.$$

Notation 0.8. We denote $((\mathrm{Id}_E)_{i=1}^n)_\# \mu \in \mathrm{Marg}((\mu)_{i=1}^n)$ by $\mathrm{Id}_{n,\mu}$ or simply Id_n when there is no ambiguity. With $n = 2$, $\mathrm{Id}_{2,\mu} = \mathrm{Joint}(\mu, \mathrm{id}_E)$. It is moreover the identity transport: $\mathrm{Id}_{2,\mu} \,.\, P = P = P \,.\, \mathrm{Id}_{2,\mu'}$.

0.0.3 Action of transport plans on measures and functions

If $\mu \in \mathcal{M}(E)$ and $\mu' \in \mathcal{M}(E')$, transport plans $P \in \mathrm{Marg}(\mu, \mu')$ have an action similar to that of kernels from E to E', on (μ'-almost surely defined) classes of functions f and on measures $\theta \in \mathcal{M}(E; \mu)$ absolutely continuous with respect to μ. For instance, the latter are transported in $\mathcal{M}(E', \mu')$, as follows:

$$\text{if } \theta \ll \mu \text{ has density } g \text{ and } B' \in \mathcal{B}(E'), \quad (\theta \,.\, P)(B') = \int_{B'} \int_{E} g(x) \mathrm{d}P(x, y).$$

If k is a kernel from E to E', $\theta \,.\, k = \theta \,.\, \mathrm{Joint}(\mu, k)$. Conversely $\theta \,.\, P = \theta \,.\, k_P$.

Definition 0.9. Take $P \in \mathrm{Marg}(\mu_1, \ldots, \mu_k)$, or $P \in \mathrm{Marg}((\mu_t)_{t\in\mathbb{R}})$. We define its transpose ${}^{\mathrm{t}}P \in \mathrm{Marg}(\mu_k, \ldots, \mu_1)$, respectively ${}^{\mathrm{t}}P \in \mathrm{Marg}((\mu_{-t})_{t\in\mathbb{R}})$ by: ${}^{\mathrm{t}}P(B_1 \times \ldots \times B_k) = P(B_k \times \ldots \times B_1)$, respectively ${}^{\mathrm{t}}P(\prod_i B_{t_i}) = P(\prod_i B_{-t_i})$.

The notation ${}^{\mathrm{t}}P$ is a reference to a transposed matrix in Remark 0.5. For $P = \mathrm{Marg}(\mu, \nu)$ we will often consider the bilinear map

$$B : (\theta, f) \longmapsto (\theta P) f = \theta(Pf) = \iint g(x) f(y)\, P(\mathrm{d}x, \mathrm{d}y),$$

where g is the density of θ with respect to μ. The case $\theta = \mu\lfloor_{]-\infty,x]}$, $f = \mathbb{1}_{]-\infty,y]}$ with $B(\theta, \nu) = P(]-\infty, x] \times]-\infty, y])$ is of special interest, see Section 3.1.

0.0.4 Concatenation of transport plans

(See, e.g., [28, p. 111], [46, p. 23].)

Definition 0.10. If $\mu_i \in \mathcal{P}(E_i)$ for $i \in \{1, 2, 3\}$, if $P_{1,2} \in \mathrm{Marg}(\mu_1, \mu_2)$ and $P_{2,3} \in \mathrm{Marg}(\mu_2, \mu_3)$, their concatenation $P_{1,2} \circ P_{2,3}$ is the unique $R \in \mathcal{P}(\mathbb{R}^3)$ such that for every $(B_1, B_2, B_3) \in \mathcal{B}(E_1) \times \mathcal{B}(E_2) \times \mathcal{B}(E_3)$:

$$R(B_1 \times B_2 \times B_3) = \int_{x\in B_1} \int_{y\in B_2} \int_{z\in B_3} \mathrm{d}\mu_1(x) k_{1,2}(x, \mathrm{d}y) k_{2,3}(y, \mathrm{d}z). \quad (1)$$

In particular, $R \in \mathrm{Marg}((\mu_1, \mu_2, \mu_3)$, $(\mathrm{proj}^{1,2})_{\#}R = P_{1,2}$, and $(\mathrm{proj}^{2,3})_{\#}R = P_{2,3}$.

Remark 0.11. Let $k_{i,j}$ be a disintegration kernel for $P_{i,j}$ and $P_{2,1} \coloneqq {}^{\mathrm{t}}P_{1,2}$. The right side of (1) also reads: $\iint_{B_1\times B_2} \int_{B_3} k(y, \mathrm{d}z) \mathrm{d}P_{1,2}(x, y)$, hence also:

$$\int_{y\in B_2} \iint_{x\in B_1,\, z\in B_3} k_{2,1}(y, \mathrm{d}x) k_{2,3}(y, \mathrm{d}z) \mu_2(\mathrm{d}y). \quad (2)$$

Concatenation is "reversed when time is reversed" in the sense that ${}^{\mathrm{t}}P_{2,3} \circ {}^{\mathrm{t}}P_{1,2} = {}^{\mathrm{t}}(P_{1,2} \circ P_{2,3})$; this is immediate after (2). Formula (1) gives immediately that $\circ$ is associative, leading to its following generalization:

$$P_{1,2} \circ \ldots \circ P_{n-1,n}\Big(\prod_{i=1}^{n} B_i\Big) = \int_{(x_i)_i \in \prod_i B_i} \mathrm{d}\mu_1(x_1) k_{1,2}(x_1, \mathrm{d}x_2) \ldots k_{n-1,n}(x_{n-1}, \mathrm{d}x_n).$$

Remark 0.12. a. In Section 0.0.2, one can also define $P.Q$ as $\mathrm{proj}_{\#}^{E,E''}(P \circ Q)$.

b. The composition and concatenation of transport plans find an easy interpretation in terms of random variables. If (X_1, X_2, X_3) is a random vector with $\mathrm{Law}(X_1, X_2) = P$ and $\mathrm{Law}(X_2, X_3) = Q$ such that $(X_i)_{i\in\{1,2,3\}}$ is a Markov process (X_1 and X_3 are independent conditionally on X_2), then $P.Q$ is the law of (X_1, X_3) and $P \circ Q$ the law of (X_1, X_2, X_3), see (2).

0.0.5 The Markov Property

We introduce the Markov property here in an alternative, equivalent way to the usual one.

Remark/Definition 0.13. As recalled in (4), a process $(X_t)_{t\in\mathbb{R}}$ is said to be *Markov* if:

$$\forall\, s \in \mathbb{R}, \forall\, t > s, \quad \mathrm{Law}(X_t |\, (X_u)_{u\leqslant s}) = \mathrm{Law}(X_t |\, X_s).$$

Denoting $\mathrm{Law}((X_t)_t) \in \mathcal{P}(\mathbb{R}^{\mathbb{R}})$ by P, this is equivalent to the fact that for all finite subset $S = \{s_1, \dots, s_d\}$ of $\mathbb{R}$, $(\mathrm{proj}^S)_{\#}P$ is the concatenation $P^{s_1,s_2} \circ \cdots \circ P^{s_{d-1},s_d}$, where $P^{s_i,s_{i+1}}$ denotes $(\mathrm{proj}^{\{s_i,s_{i+1}\}})_{\#}P$. More generally, we say that any measure $P \in \mathcal{P}(\mathbb{R}^{\mathbb{R}})$ is Markov if it satisfies this property.

We extend these definition in the obvious way to processes indexed on subsets $R \subset \mathbb{R}$ and measures on $\mathcal{P}(\mathbb{R}^R)$.

We recall the Kolmogorov–Daniell theorem and its usual corollary on Markov measures.

Proposition 0.14 (Kolmogorov–Daniell theorem)**.** *Let E be a Polish space and, for each finite subset S of $\mathbb{R}$, μ_S be a probability measure on E^S. If $(\mathrm{proj}^{S'})_{\#}\mu_S = \mu_{S'}$ for any $S' \subset S$, there exists a unique $P \in \mathcal{P}(E^{\mathbb{R}})$ with $(\mathrm{proj}^S)_{\#}P = \mu_S$ for all S.*

One of the most usual applications of Proposition 0.14 is for measures μ_S of type $\mu_{s_1,s_2} \circ \cdots \circ \mu_{s_{d-1},s_d}$, where $S = \{s_1, \dots, s_d\}$.

Corollary 0.15. *Let $(\mu_{s,t})_{s<t}$ be a family of transport plans in $\mathcal{P}(E \times E)$ such that:*

$$\mu_{s,u} = \mu_{s,t} \cdot \mu_{t,u}$$

for every $s < t < u$. Then there exists a unique Markov measure $P \in \mathcal{P}(E^{\mathbb{R}})$ with $P^{s,t} = \mu_{s,t}$ for every $s < t$.

Definition 0.16. It is usual to call *consistent family* every family $(\mu_S)_S$ or $(\mu_{s,t})_{s<t}$ as in Proposition 0.14 and Corollary 0.15.

Theorem 0.17 ([28, Th. 1])**.** *Let $(\mu_t)_{t\in\mathbb{R}}$ be a family of probability measures on some Polish space E, and for every $s < t$ let $\mathcal{N}_{s,t} \subset \mathcal{P}(E^2)$ be a set of transport plans. Assume that:*

i. for every s, t, $\mathcal{N}_{s,t}$ is not empty,

ii. for every s, t, $\mathcal{N}_{s,t} \subset \mathrm{Marg}(\mu_s, \mu_t)$,

iii. for every s,t, $\mathcal{N}_{s,t}$ is closed for the weak topology,

iv. for $r<s<t$ and any $(P,P')\in\mathcal{N}_{r,s}\times\mathcal{N}_{s,t}$, $P\,.\,P'\in\mathcal{N}_{r,t}$,

v. for every d and $t_1<\cdots<t_d$, if the sequences $(Q^n_{t_i,t_{i+1}})_n\in\mathcal{N}_{t_i,t_{i+1}}$ converge weakly to $Q_{t_i,t_{i+1}}$, then the sequence $(Q^n_{t_1,t_2}\circ\cdots\circ Q^n_{t_{d-1},t_d})_n$ tends weakly to $Q_{t_1,t_2}\circ\cdots\circ Q_{t_{d-1},t_d}$.

Then, there exists a Markov measure $P\in\mathrm{Marg}((\mu_t)_t)$ with $(\mathrm{proj}^{s,t})_\#P\in\mathcal{N}_{s,t}$ for every $s<t$.

Notation 0.18. a. Let us denote $(x_i)_{i=1}^d$ and $(y_i)_{i=1}^d$ in $\mathbb{R}^d$ by x and y. We endow $\mathbb{R}^d$ with the natural partial order defined by:

$$x\leqslant y\quad\text{if:}\quad\forall\, i, x_i\leqslant y_i.$$

We also set $[x,y]:=\{z\in\mathbb{R}^d : x\leqslant z\leqslant y\}=\prod_i[x_i,y_i]$ and similarly $]x,y]$, etc. In particular $]-\infty,x]=]-\infty,x_1]\times\cdots\times]-\infty,x_d]$.

b. Several times appear statements where some intervals have to be considered closed or open at some of their bounds, either arbitrarily or depending on possible cases. To alleviate the writing, we introduce the symbol "$\int$" and place it at these bounds.

Definition 0.19. If $\mu\in\mathcal{M}(\mathbb{R}^d)$, its cumulative distribution function F_μ, that we also denote by $F[\mu]$ to avoid multiple subscripts, is defined, using Notation 0.18, by:

$$F_\mu : x\in\mathbb{R}^d\longmapsto\mu(]-\infty,x]).$$

Reminder 0.20. Recall for instance from [25, Th. 3.25] that such functions F are characterized by the fact that:

a. for the natural partial order of $\mathbb{R}^d$ (see Notation 0.18), F is increasing and upper semi-continuous in the sense that for all $x\in\mathbb{R}^d$:

$$\forall\,\varepsilon>0,\,\exists\,\eta>0,\,\forall\, y\geqslant x,\quad \|y-x\|_\infty\leqslant\eta\implies F(x)\leqslant F(y)\leqslant F(x)+\varepsilon. \tag{3}$$

b. $\lim_{\min_i(x_i)\to-\infty}F(x)=0$ and $\lim_{\min_i(x_i)\to+\infty}F(x)=1$,

c. for every $h=(h_1,\dots,h_d)\in[0,+\infty[^d$ and $x\in\mathbb{R}^d$, the quantity

$$\sum_\varepsilon\sigma(\varepsilon)F(x+\varepsilon h),$$

which is the measure of the rectangle $]x,x+h]\subset\mathbb{R}^d$, is non-negative. Here $\varepsilon=(\varepsilon_1,\dots,\varepsilon_d)$ ranges over $\{0,1\}^d$, σ is 1 if $\sum\varepsilon_i$ is even, -1 otherwise, and εh means $(\varepsilon_1h_1,\dots,\varepsilon_dh_d)$.

Definition 0.21. If $d\in\mathbb{N}^*$ and $m\in]0,+\infty[$, following [43, §6.G], we define the *lower orthant order* on $\{\mu\in\mathcal{M}(\mathbb{R}^d):\mu(\mathbb{R}^d)=m\}$ by: $\mu\preceq_{\mathrm{lo}}\nu$ if $F_\mu\geqslant F_\nu$.

Definition 0.22. We call lower orthant supremum of a family $(P_\tau)_{\tau\in\mathcal{T}}$ of measures of same mass m on $\mathbb{R}^d$, the smallest upper bound of $\{P_\tau\}_\tau$ for $\preceq_{\mathrm{lo}}$, if it exists, i.e. a measure P of mass m such that:

- for every τ, $P_\tau \preceq_{\text{lo}} P$,
- $P \preceq_{\text{lo}} Q$ as soon as $P_\tau \preceq_{\text{lo}} Q$ for every τ.

By definition, if it exists it is unique. We denote it by $\operatorname{losup}_\tau P_\tau$. Similarly we define $\operatorname{loinf}_\tau P_\tau$.

For every increasing $h : \mathbb{R} \to [0,1]$, $P \,.\, h$ is μ-almost surely increasing, i.e. more exactly, there is an increasing $\tilde{h} : \mathbb{R} \to [0,1]$ such that, for every bounded continuous function g:

$$\int g(x)h(y)\mathrm{d}P(x,y) = \int g(x)\tilde{h}(x)\mathrm{d}\mu(x).$$

Notation 0.23. If $\mu \in \mathcal{M}(\mathbb{R})$ we denote by $\mathcal{M}(\mu)$ and $\mathcal{P}(\mu)$ the sets of positive measures, respectively probability measures, absolutely continuous with respect to μ, and $\mathcal{M}^\searrow(\mu)$ and $\mathcal{P}^\searrow(\mu)$, or $\mathcal{M}^\searrow$ and $\mathcal{P}^\searrow$ if there is no ambiguity, their subsets of measures with bounded and decreasing density.

Definition 0.24. Take $\mu \in \mathcal{P}(\mathbb{R})$. The *quantile function*

$$G_\mu : [0,1] \to \mathbb{R} \cup \{-\infty, +\infty\}$$

is the increasing left-continuous function such that $(G_\mu)_\# \lambda|_{[0,1]} = \mu$, i.e. the generalized inverse of the cumulative distribution function F_μ:

$$G_\mu(q) = \inf\{x \in \mathbb{R} : \, F_\mu(x) \geq q\}.$$

The quantile measure $\mathfrak{Q}((\mu_\tau)_{\tau\in\mathcal{T}})$ is obtained by pushing forward $\lambda|_{[0,1]}$ on $\mathcal{P}(\mathbb{R}^\mathcal{T})$ by the map $G = (G_{\mu_\tau})_{\tau\in\mathcal{T}} : x \in [0,1] \mapsto (G_{\mu_\tau}(x))_{\tau\in\mathcal{T}} \in \mathbb{R}^\mathcal{T}$. In other words, the functions G_{μ_τ} can be seen as random variables defined on $([0,1], \lambda|_{[0,1]})$ and $\mathfrak{Q}((\mu_\tau)_{\tau\in\mathcal{T}})$ is the law of the process $(G_{\mu_\tau})_{\tau\in\mathcal{T}}$. One can check from this definition the equality in (8) and (10).

Definition 0.25. If $M \in \operatorname{Marg}((\mu_t)_t)$ and if $R = \{r_1, \dots, r_m\} \subset \mathbb{R}$, we denote by $M_{[R]} \in \operatorname{Marg}((\mu_t)_{\in\mathbb{R}})$ the measure M *made Markov at the points of* R defined by the data of its finite marginals $(\operatorname{proj}^S)_\# M_{[R]}$, for all finite S containing R, as follows:

$$(\operatorname{proj}^S)_\# M_{[R]} = M^{s_1^0,\dots,s_{n_0}^0,r_1} \circ M^{r_1,s_1^1,\dots,s_{n_1}^1,r_2} \circ \dots \circ M^{r_m,s_1^m,\dots,s_{n_m}^m},$$

where $S = \{s_1^0, \dots, s_{n_0}^0, r_1, s_1^1, \dots, s_{n_1}^1, r_2, \dots, r_m, s_1^m, \dots, s_{n_m}^m\}$ and where the first or last term disappears if n_0 or n_m is null, respectively. These marginals are consistent in the sense of Definition 0.16. So by Proposition 0.14 this defines $M_{[R]}$. We also commit an abuse of language: $M_{[R]}$ is rather the "law of a process X of law M, made Markov at the points of R".

By Proposition 0.63, for any $R \subset \mathbb{R}$, $(L_{R\cap[s,t]})_{s<t} \in \operatorname{Marg}(\lambda, \lambda)$ is also a consistent family, thus again Proposition 0.14 enables us to define the following processes on the set of quantile levels.

Definition 0.26. For all $R \subset \mathbb{R}$ we denote by $\operatorname{Lev}_R \in \operatorname{Marg}((\lambda_t)_{t\in\mathbb{R}})$ (λ_t denotes λ at each t) the Markov process with 2-marginals $\operatorname{Lev}_R^{s,t} = L_{R\cap[s,t]}$. We call $\operatorname{Lev}_\mathbb{R}$ the *level process* attached to $(\mu_t)_{t\in\mathbb{R}}$ and denote it by Lev.

Remark 0.27. In this subsection, using Definition 0.25 with $M = \mathfrak{Q}$ we obtain measures $\mathfrak{Q}_{[R]}$ linked with Lev_R as follows. After Remark 0.66, $\mathfrak{Q}^{s,t}_{[R]} = (G_s \otimes G_t)_\# L_{[s,t]\cap R}$, thus, with $G = (\otimes_{t\in\mathbb{R}} G_t)$, the measures $G_\#\mathrm{Lev}_R$ and $\mathfrak{Q}_{[R]}$ have the same 2-marginals. They are actually equal, as we prove in the "claim" at the beginning of the proof of Theorem 0.67(b).

$$\forall\, s \in \mathbb{R},\, \forall\, t > s, \quad \mathrm{Law}(X_t |\, (X_u)_{u\leqslant s}) = \mathrm{Law}(X_t |\, X_s), \tag{4}$$

where $\mathrm{Law}(X_t |\, (X_u)_{u\leq s})$ is the law of X_t conditionally to the σ-algebra generated by the X_u for $u \leq s$. See also Definition 0.13, where the Markov property is introduced only through notions defined in this article.

Theorem A. *There is an increasing sequence $(R_n)_{n\in\mathbb{N}}$ of finite subsets of $\mathbb{R}$ such that $\mathfrak{Q}_{[R_n]} \in \mathrm{Marg}((\mu_t)_{t\in\mathbb{R}})$ converges weakly to $\mathfrak{MQ}$.*

Lemma 0.28 (Continuity of $\circ$ when the kernels are increasing)**.** *Take $(\mu_1, \dots, \mu_n) \in \mathcal{P}(\mathbb{R})^n$ and for all $i \in [\![1, n-1]\!]$ a closed set $\mathfrak{I}_{i,i+1} \subset \mathrm{Marg}(\mu_i, \mu_{i+1})$ of transport plans with increasing kernel. The sets $\mathfrak{I}_{i,i+1}$ satisfy Property* (v) *stated for the sets $\mathcal{N}_{t_i,t_{i+1}}$ in Theorem 0.17.*

Lemma 0.29. *Take μ and μ' in $\mathcal{P}(\mathbb{R})$. The space of transport plans with increasing kernel in $\mathrm{Marg}(\mu, \mu')$ is closed for the weak topology.*

Remark 0.30.

a. In Reminder 0.20, (a) and the first limit of (b) pass to the infimum of functions that are both monotone and upper semi-continuous. If moreover $(P_\tau)_\tau$ has an upper bound P, then the second limit of (b) holds. Indeed, for all τ, $F[P_\tau] \geqslant F[P]$, so that $\inf_\tau F[P_\tau] \geqslant F[P]$, and besides $\lim_{\min_i(x_i)\to+\infty} F[P](x) = 1$. Thus, if $(P_\tau)_\tau$ is bounded from above, $\inf_\tau F[P_\tau]$ satisfies (a)–(b) of Reminder 0.20, so is a cumulative distribution function if and only if it satisfies (c).

b. If (c) of Reminder 0.20 is satisfied by the functions $F[P_\tau]$, then $(P_\tau)_\tau$ has a lower orthant supremum, and $F[\mathrm{losup}_\tau P_\tau] = \inf_\tau F[P_\tau]$.

Remark/Notation 0.31. If $d = 1$ the order $\preceq_{\mathrm{lo}}$ is usually called stochastic order and denoted by $\preceq_{\mathrm{sto}}$; we will then call "stochastic supremum" the lower orthant supremum of Definition 0.22 and denote it by stosup.

Lemma 0.32 (Existence criteria for losup)**.**

a. If, for $\preceq_{\mathrm{lo}}$, a sequence $(P_n)_{n\in\mathbb{N}} \in (\mathcal{P}(\mathbb{R}^d))^{\mathbb{N}}$ is bounded from above, and increasing, i.e. $n \leqslant m \Rightarrow P_n \preceq_{\mathrm{lo}} P_m$, then $\mathrm{losup}_n P_n$ exists and $(P_n)_n$ converges weakly to it.

b. If a family $(P_\tau)_{\tau\in\mathcal{T}}$ is bounded from above for $\preceq_{\mathrm{lo}}$ and if for every $\tau, \tau' \in \mathcal{T}$ there exists $\sigma \in \mathcal{T}$ such that $P_\sigma \succeq_{\mathrm{lo}} P_\tau$ and $P_\sigma \succeq_{\mathrm{lo}} P_{\tau'}$, then $\mathrm{losup}_\tau P_\tau$ exists and there is an increasing sequence $(P_{\tau_n})_n$ that converges weakly to it.

The results extend in an obvious way to measures of mass $m > 0$ in $\mathcal{M}(\mathbb{R})$.

Proof. (a) Set $F := \inf_n F[P_n]$. After Remark 0.30, showing that F satisfies (c) of Reminder 0.20 ensures the existence of $\mathrm{losup}_n P_n$. Consider $M := \sum_\varepsilon \sigma(\varepsilon) F(x + \varepsilon h)$ as in Reminder 0.20(c). Since $F[P_n]$ is decreasing, F is its

simple limit, so M is the simple limit of $\sum_\varepsilon \sigma(\varepsilon)F[P_n](x+\varepsilon h)$, hence $M \geqslant 0$. We are done. The weak convergence is given by the pointwise convergence of the cumulative distribution functions, see Reminder 0.42.

(b) This is a diagonal construction. Let $\mathcal{C} = \bigcup_{k\in\mathbb{N}}\{x_k\}$ be a countable dense set in $\mathbb{R}^d$. Set $F \coloneqq \inf_{\tau\in\mathcal{T}} F_\tau$. Then for every $(k,n) \in \mathbb{N}^{*2}$ we find $\tau_{k,n} \in \mathcal{T}$ such that $F[P_{\tau_{k,n}}](x_k) \leqslant F(x_k) + 1/n$. For each n, by a finite induction using the assumption of (b) on the $P_{\tau_{k,n}}$, we find $\sigma_n \in \mathcal{T}$ such that: $\forall\, k \leqslant n$, $F[P_{\sigma_n}](x_k) \leqslant F(x_k) + 1/n$ and $(P_{\sigma_n})_n$ is increasing. Hence:

$$\forall\, x \in \mathcal{C}, \quad F[P_{\sigma_n}](x) \to F(x). \tag{5}$$

By (a), $P = \operatorname{losup}_n P_{\sigma_n}$ exists and:

$$\forall\, x \in \mathbb{R}^d, \quad F[P_{\sigma_n}](x) \to F[P](x). \tag{6}$$

Let us prove that (5) holds for any $x \in \mathbb{R}^d$, so that $F = F[P]$. Assume by contradiction that, for some x, $F(x) < \inf_n F[P_{\sigma_n}](x)$, i.e. by definition of F, that for some $\tau \in \mathcal{T}$, $F(x) < F[P_\tau](x) < \inf_n F[P_{\sigma_n}](x) = F[P](x)$. Now $F[P_\tau]$ is upper semi-continuous so in a neighbourhood $\mathcal{U}$ of x in $[x, +\infty[$, $F[P_\tau] < F[P]$. But on the dense set $\mathcal{C}$, by (5) and (6), $F[P] = F$, hence on $\mathcal{U} \cap \mathcal{C} \neq \varnothing$, $F[P_\tau] < F$, a contradiction.

Finally $F = F[P]$, so F is a cumulative distribution function, so by Remark 0.30(b), $P = \operatorname{losup}_{\tau\in\mathcal{T}} P_\tau$. Moreover, $P_{\sigma_n} \to P$. □

(3.11(b)) $\Rightarrow$ (3.11(a)): if $\theta \preceq_{\text{sto}} \theta'$, take $h : \mathbb{R} \to [0,1]$ increasing, then $(\theta \,.\, P) \,.\, h = \theta \,.\, \tilde{h} \leq \theta' \,.\, \tilde{h}$ since $\theta \preceq_{\text{sto}} \theta'$ and $\tilde{h}$ is increasing by (3.11(b)). Then $\theta' \,.\, \tilde{h} = (\theta' \,.\, P) \,.\, h$ yields (3.11(a)).

Remark 0.33. A direct consequence of the first point of Proposition 3.11 is that, if $P \in \operatorname{Marg}(\mu_1, \mu_2)$ and $Q \in \operatorname{Marg}(\mu_2, \mu_3)$ have increasing kernel, so has the result $P \,.\, Q$ of their composition.

Remark 0.34 ($\mathcal{M}^{\searrow}(\mu)$ is closed for the weak topology)**.** A measure θ belongs to $\mathcal{M}^{\searrow}(\mu)$ if and only if its density is bounded and, for all $(a,b,c,d) \in \mathbb{R}^4$:

$$\big(a < b < c < d \text{ and } \mu([a,b]) \,.\, \mu([c,d]) \neq 0\big) \implies \frac{\theta([a,b])}{\mu([a,b])} \geqslant \frac{\theta([c,d])}{\mu([c,d])}. \tag{7}$$

"Only if" is clear. For the "if" part, by the Lebesgue differentiation theorem,

$$x \longmapsto \lim_{\varepsilon\to 0} \frac{\theta([x-\varepsilon, x+\varepsilon])}{\mu([x-\varepsilon, x+\varepsilon])}$$

provides a representative of the density. Now if a sequence $(\theta_n)_n$ satisfies (7) and weakly tends (see Reminder 0.42) to $\theta \in \mathcal{M}(\mu)$, θ satisfies it also (if a, b, c or d is an atom of θ, re-obtain (7) by limit of larger intervals).

Remark 0.35 (P has increasing kernel if and only if ${}^{\mathrm{t}}P$ maps $\mathcal{M}^{\searrow}(\nu)$ to $\mathcal{M}^{\searrow}(\mu)$)**.** In Proposition 3.11, (b) is equivalent to the same statement with decreasing functions; in turn, transposing, this means that, for all decreasing $h : \mathbb{R} \to [0,1]$, $(h\nu) \,.\, {}^{\mathrm{t}}P$, which is equal to $(P \,.\, h) \,.\, \mu$, has decreasing density.

0.1 Quantile measures and minimal couplings

We define the quantile coupling (Definition 0.37) and quantile process law (Definition 0.39) through a minimality property that is crucial in our paper. We also state the direct and more classical Definition 0.24 of the quantile measure. Further characterizations of these coupling and process are given throughout the paper and in [8] where the approach is optimal transport. The reader can refer to [41, 45] for more background. See also the papers [23, 40].

Reminder 0.36. Take μ and ν in $\mathcal{M}(\mathbb{R})$. Every P of $\mathrm{Marg}(\mu,\nu)$ satisfies the Fréchet–Hoeffding bound:

$$\forall\,(x,y)\in\mathbb{R}^2,\quad F_P(x,y)\le\min(F_\mu(x),F_\nu(y)). \tag{8}$$

Proposition/Definition 0.37. *There is a unique P such that* (8) *is an equality. We call it the Fréchet–Hoeffding, comonotonic or* quantile coupling *and denote it by $\mathfrak{Q}(\mu,\nu)$. Said briefly: $\mathfrak{Q}(\mu,\nu)=\mathrm{loinf}\,\mathrm{Marg}(\mu,\nu)$.*

The proof follows from Definition 0.24, see below.

Remark 0.38 (Minimality in the language of transport plans)**.** a. The quantile process $\mathfrak{Q}$ may also be defined by the fact that its transitions are *minimal* among those of all transport plans of $\mathrm{Marg}(\mu,\nu)$: for any $P\in\mathrm{Marg}(\mu,\nu)$ and fixed x, $F[\mu\lfloor_{]-\infty,x]}\,.\,k_P](y)=F_P(x,y)$ for every $y\in\mathbb{R}$, hence, after Definition 0.37 and the characterization of stochastic order of Remark 3.9:

$$\forall\,P\in\mathrm{Marg}(\mu,\nu),\quad \mu\lfloor_{]-\infty,x]}\,.\,\mathfrak{Q}(\mu,\nu)\preceq_{\mathrm{sto}}\mu\lfloor_{]-\infty,x]}\,.\,k_P \tag{9}$$

(in fact, $\mu\lfloor_{]-\infty,x]}\,.\,\mathfrak{Q}(\mu,\nu)$ is of the type $\nu\lfloor_{]-\infty,y[}+a\delta_y$), i.e. the quantile coupling maps the measures $\mu\lfloor_{]-\infty,x]}$ to the stochastically smallest possible measures, (9) being also true for any $\theta\in\mathcal{M}^{\searrow}(\mu)$ in place of $\mu\lfloor_{]-\infty,x]}$.

Notice that a property similar to (9), with $\preceq_C$ in place of $\preceq_{\mathrm{sto}}$, defines the (left-)curtain coupling in [4].

b. As $\mu\lfloor_{]x,+\infty[}\,.\,k_P=\nu-\mu\lfloor_{]-\infty,x]}\,.\,k_P$, $\mu\lfloor_{]x,+\infty[}\,.\,\mathfrak{Q}(\mu,\nu)$ is paradoxically *maximal* for $\preceq_{\mathrm{sto}}$, hence minimal transitions mean that the mass of μ is mixed as less as possible when transported on that of ν.

Proposition/Definition 0.39. *If $(\mu_\tau)_{\tau\in\mathcal{T}}$ is a family of measures, there is a unique measure $\mathfrak{Q}\in\mathrm{Marg}((\mu_\tau)_{\tau\in\mathcal{T}})$ such that for every $\tau\neq\sigma$, the transport plan $\mathfrak{Q}^{\tau,\sigma}=(\mathrm{proj}^{\{\tau,\sigma\}})_{\#}\mathfrak{Q}$ is the quantile coupling $\mathfrak{Q}(\mu_\tau,\mu_\sigma)$. We call it the* quantile measure *of $(\mu_\tau)_{\tau\in\mathcal{T}}$.*

Proof of Propositions 0.37 and 0.39. The existence parts follows from Definition 0.24 below and are proved just after it; in Proposition 0.37 uniqueness is clear; let us prove it in Proposition 0.39. It is rather easy to prove that the equality in (8) for every pair of measures implies the equality in the Fréchet–Hoeffding bound of general dimension:

$$\forall\,(x_i)_{i=1}^d\in\mathbb{R}^d,\quad F_{P^S}(x_1,\dots,x_d)\le\min_{i\in\{1\dots,d\}}(F_{\mu_{s_i}}(x_i)), \tag{10}$$

where $S = \{s_1, \dots, s_d\}$ is any finite subset of $\mathbb{R}$. (Take j such that $F_{\mu_{s_j}}(x_j) = \min_{1\le i\le d}(F_{\mu_{s_i}}(x_i))$. For every $k \neq j$,

$$F[\mathrm{proj}_{\#}^{\{s_j,s_k\}} P^S](x_j, x_k) = F[\mu_{s_j}](x_j)$$

so that finally $P^S(]-\infty, x]) = F[\mu_{s_j}](x_j)$.) Since equalities (10) for all finite $S \subset \mathbb{R}$ are a compatible set of conditions, this proves with Proposition 0.14 that there exists at most one quantile measure P in $\mathrm{Marg}((\mu_\tau)_{\tau\in\mathcal{T}})$. □

Notation 0.40. We denote by $\mathfrak{Q}(\mu_{s_1}, \dots, \mu_{s_d})$ and $\mathfrak{Q}((\mu_t)_{t\in\mathbb{R}})$ the multidimensional quantile coupling and the quantile measure.

Remark 0.41 (Products of quantile couplings have increasing kernel and map $\mathcal{M}^{\searrow}$ on $\mathcal{M}^{\searrow}$)**.** We will see in Section 0.3 that $\mathfrak{Q}(\mu, \nu)$ is a composition of two transport plans with increasing kernel. Therefore by Remark 0.33 it has increasing kernel. Since ${}^{t}\mathfrak{Q}(\mu, \nu) = \mathfrak{Q}(\nu, \mu)$, Remark 0.35 ensures that $\mathfrak{Q}(\nu, \mu)$ maps $\mathcal{M}^{\searrow}(\mu)$ to $\mathcal{M}^{\searrow}(\nu)$.

All this also ensure both properties for products of quantile couplings.

0.2 Distances ρ and $\tilde{\rho}$

0.2.1 A distance that metricizes $\mathrm{Marg}(\mu, \nu)$

Reminder 0.42. We remind the reader of the "Portmanteau theorem" (see, e.g., [7, Th. 2.1]): the weak convergence on some metric space E endowed with its Borel σ-algebra is (equivalently) defined by:

$$P_n \underset{n\to\infty}{\longrightarrow} P \text{ if } P_n(R) \underset{n\to\infty}{\longrightarrow} P(R) \text{ for all } R \text{ such that } P(\partial R) = 0. \qquad (11)$$

In $\mathbb{R}^n$, it is equivalent to consider in (11) only sets R of the form $\prod_{i=1}^n]-\infty, x_i]$, see [7, Exam. 2.3 p. 18].

Proposition/Notation 0.43. *Let ρ be the function defined on $\mathcal{P}(\mathbb{R}^d)^2$ by:*

$$\rho(P, Q) = \|F_P - F_Q\|_\infty.$$

(See Definition 0.19 for F_P and F_Q.) This is a distance. If $\mu_1, \dots, \mu_d$ are probability measures on $\mathbb{R}$, it induces the weak topology on $\mathrm{Marg}((\mu_i)_{1\le i\le d})$. More precisely, for any $P \in \mathrm{Marg}((\mu_i)_{1\le i\le d})$ and any sequence $(P_n)_{n\in\mathbb{N}}$ of elements of $\mathrm{Marg}((\mu_i)_{1\le i\le d})$, the following are equivalent:

i. For all $x = (x_i)_{i=1}^d$, $P\big(\partial\big(\prod_{i=1}^d]-\infty, x_i]\big)\big) = 0 \Rightarrow \lim_n F_{P_n}(x) = F_P(x)$,

ii. For all $x = (x_i)_{i=1}^d$, $\lim_n F_{P_n}(x) = F_P(x)$,

ii'. For all $x = (x_i)_{i=1}^d$, $\lim_n \big(P_n\big(\prod_i]-\infty, x_i\rfloor\big)\big) = P\big(\prod_i]-\infty, x_i\rfloor\big)$,

iii. $(F_{P_n})_{n\in\mathbb{N}}$ converges uniformly to F_P.

Proof. The fact that ρ is a distance is immediate. To get (iii) ⇒ (ii') use that $]-\infty, x[= \bigcup_{y<x}]-\infty, y]$. Now it suffices to prove (i) ⇒ (ii) ⇒ (iii).

(i) ⇒ (ii). Take $Q \in \mathrm{Marg}((\mu_i)_{1\le i\le d})$ and $b = (b_i)_{i=1}^d$. For $b' = (b'_i)_{i=1}^d \geq b$:

$$F_Q(b') - F_Q(b) = Q(R') - Q(R) \leq \mu_1(]b_1, b'_1]) + \cdots + \mu_d(]b_d, b'_d]), \qquad (12)$$

where $R' =]-\infty, b']$ and $R =]-\infty, b]$, which shows that $(F_Q)_Q$ is "equicontinuous on the right". Take $\varepsilon > 0$. Since each $b_i \in \mathbb{R}$ can be approached from the right by a sequence of non-atomic points for μ_i, there exists $b' \geqslant b$ such that

$$\mu_1(]b_1, b_1']) + \cdots + \mu_d(]b_d, b_d']) < \varepsilon \quad \text{and} \quad \mu_1(b_1') + \cdots + \mu_d(b_d') = 0,$$

hence in particular $Q(\partial R') = 0$ for every $Q \in \mathrm{Marg}((\mu_i)_{1\leq i\leq d})$ so that (i) applies to R' for P and all the P_n. Thus:

$$\begin{aligned} |P_n(R) - P(R)| &\leq |P_n(R) - P_n(R')| + |P_n(R') - P(R')| + |P(R') - P(R)| \\ &\leq 2\varepsilon + |P_n(R') - P(R')|, \end{aligned}$$

and if (i) holds, $|P_n(R') - P(R')| \leqslant \varepsilon$ for n great enough, hence (ii) follows.

(ii) $\Rightarrow$ (iii). We apply an adapted version of the prior argument. Suppose (ii) and take $\varepsilon > 0$. For every i there exists a finite sequence

$$-\infty = b_0^{(i)}(\varepsilon) < \cdots < b_{N(\varepsilon,i)}^{(i)}(\varepsilon) = +\infty$$

avoiding the big atoms of μ_i, i.e. so that $\mu_i(]b_k^{(i)}, b_{k+1}^{(i)}[) < \varepsilon$ (this is classical and is proved, e.g., in [7, §12], which deals with the modulus of continuity of càdlàg paths). Every $R =]-\infty, b]$ contains some rectangle R_ε^- and is included in the interior of some rectangle R_ε^+, both bounded by consecutive points $(b_k^{(i)})_{i,k}$. Using again (12) for the first and last terms:

$$|P_n(R) - P(R)| \leq \underbrace{|P_n(R) - P_n(R_\varepsilon^-)|}_{\leqslant d\varepsilon} + \underbrace{|P_n(R_\varepsilon^-) - P(R_\varepsilon^-)|}_{\substack{\longrightarrow \\ n\to\infty} 0 \text{ by (ii)} \quad (*)} + \underbrace{|P(R_\varepsilon^-) - P(R)|}_{\leqslant d\varepsilon},$$

where $(*)$ is uniform as the rectangles R_ε^- are finitely many. We get (iii). □

0.2.2 Another expression for ρ; an alternative distance $\tilde{\rho}$

Let $\mu_1, \ldots, \mu_d$ be probability measures on $\mathbb{R}$ and P, Q stand for elements of $\mathrm{Marg}((\mu_i)_{i=1}^d)$.

Notation 0.44. When f, g and $(f_i)_{i=1}^d$ are functions, by a slight abuse in this subsection, $f.g$ denotes the function $(x, y) \mapsto f(x)g(y)$ and $\prod_{i=1}^d f_i$ the function $(x_1, \ldots, x_d) \mapsto \prod_{i=1}^d f_i(x_i)$

Remark 0.45. By definition, $\rho(P, Q) = \sup\left|\int \prod_{i=1}^d f_i \,\mathrm{d}P - \int \prod_{i=1}^d f_i \,\mathrm{d}Q\right|$, where each f_i ranges over $\{\mathbb{1}_{]-\infty,x]} : x \in \mathbb{R}\}$. But in fact:

Proposition/Definition 0.46. *a. For any P and Q in* $\mathrm{Marg}((\mu_i)_{i=1}^d)$*:*

$$\rho(P, Q) = \sup\left|\int \prod_{i=1}^d f_i \,\mathrm{d}P - \int \prod_{i=1}^d f_i \,\mathrm{d}Q\right|, \tag{13}$$

where each f_i ranges over $\{f : \mathbb{R} \to [0, 1] : f$ is decreasing$\}$.

b. Proposition 0.43 may be stated with a distance $\tilde{\rho}$ based on $x \in \mathbb{R}^n \mapsto P([x, +\infty[)$ in place of $F[P]$. Then $\tilde{\rho}$ satisfies (13) *with* increasing *functions f_i.*

Proof. Once (a) is shown, (b) is clear. To show (a), by Remark 0.45, we have only to prove $\geqslant$. For $f : \mathbb{R} \to [0,1]$ a decreasing function and $I(t) \coloneqq f^{-1}([t, +\infty[)$, $f(x) = \int_0^1 \mathbb{1}_{I(t)}(x)\mathrm{d}t$, thus for $(f_i)_{i=1}^d$ such functions:

$$\left(\prod_{i=1}^d f_i\right)(x_1, \dots, x_d) = \int_{t\in[0,1]^d} \mathbb{1}_{R(t)}(x_1, \dots, x_d)\mathrm{d}t,$$

where $R(t) = I(t_1) \times \cdots \times I(t_d)$. Therefore:

$$\begin{aligned}\left|\int \prod_{i=1}^d f_i \,\mathrm{d}P - \int \prod_{i=1}^d f_i \,\mathrm{d}Q\right| &= \left|\int_{t\in[0,1]^d} P(R(t)) - Q(R(t))\mathrm{d}t\right| \\ &\leq \int_{t\in[0,1]^d} |P(R(t)) - Q(R(t))|\mathrm{d}t \\ &\leq \rho(P, Q).\end{aligned}$$

□

Proposition 0.46 has a corollary in the case of transport plans ($d = 2$).

Proposition 0.47. *Take μ, μ' in $\mathcal{P}(\mathbb{R})$ and P, Q in $\mathrm{Marg}(\mu, \mu')$. Then:*

$$\rho(P, Q) = \sup\{\rho(\theta \,.\, P, \theta \,.\, Q) : \theta \in \mathcal{M}^{\searrow}(\mu) \text{ and } \theta \text{ has density bounded by } 1\},$$

where the distance ρ on the right of the equality is that on $\mathcal{P}(\mathbb{R})$.

Hence if, for some $\nu \in \mathcal{P}(\mathbb{R})$, $R \in \mathrm{Marg}(\nu, \mu)$ is a transport preserving $\mathcal{M}^{\searrow}$, i.e. $\forall\, \eta \in \mathcal{M}^{\searrow}(\nu)$, $\eta \,.\, R \in \mathcal{M}^{\searrow}(\mu)$, then $\rho(R \,.\, P, R \,.\, Q) \leqslant \rho(P, Q)$.

Proof. For the first equality, as $\int f \,.\, g \,\mathrm{d}P = ((f\mu) \,.\, P) \,.\, g$, by Proposition 0.46: $\rho(P, Q) = \sup_{f,g} |((f\mu) \,.\, P) \,.\, g - ((f\mu) \,.\, Q) \,.\, g|$ and $\rho(\theta \,.\, P, \theta \,.\, Q) = \sup_g |(\theta \,.\, P) \,.\, g - (\theta \,.\, Q) \,.\, g|$. Now θ is as in the proposition if and only if $\theta = f\mu$ with $f : \mathbb{R} \to [0,1]$, decreasing. The result follows. Then if $R \in \mathrm{Marg}(\nu, \mu)$ is as claimed,

$$\rho(RP, RQ) = \sup\{\rho(\theta \,.\, P, \theta \,.\, Q) |\, \theta = \bar{\theta} \,.\, R,$$

where $\bar{\theta} \in \mathcal{M}^{\searrow}(\mu)$ and $\bar{\theta}$ has density bounded by $1\}$. This set is included in that of the proposition, since the action of R on $\bar{\theta}$ does not increase the maximum of its density. □

Remark 0.48 ($\{P \in \mathrm{Marg}(\mu, \nu) : P$ maps $\mathcal{M}^{\searrow}(\mu)$ to $\mathcal{M}^{\searrow}(\nu)\}$ is closed for the weak topology)**.** Any $\theta \in \mathcal{M}^{\searrow}(\mu)$ is an increasing limit of positive combinations of characteristic functions $\mathbb{1}_{]-\infty,x]}$, so $P \in \mathrm{Marg}(\mu, \nu)$ maps $\mathcal{M}^{\searrow}(\mu)$ in $\mathcal{M}^{\searrow}(\nu)$ if and only if it maps $\{\mu\lfloor_{]-\infty,x]} \colon x \in \mathbb{R}\}$ in it. Now take a sequence $(P_n)_n \in \mathrm{Marg}(\mu, \nu)^{\mathbb{N}}$ of transport plans having this property and converging weakly to $P \in \mathrm{Marg}(\mu, \nu)$. After Proposition 0.43, $\|F[P_n] - F[P]\|_\infty \to 0$; in particular, for any $x \in \mathbb{R}$, $\|F[P_n](x, \cdot) - F[P](x, \cdot)\|_\infty \to 0$, i.e. $\|F[\mu\lfloor_{]-\infty,x]} \,.\, P_n] - F[\mu\lfloor_{]-\infty,x]} \,.\, P]\|_\infty \to 0$, i.e. $\mu\lfloor_{]-\infty,x]} \,.\, P_n$ converges weakly to $\mu\lfloor_{]-\infty,x]} \,.\, P$. By Remark 0.34, $\mu\lfloor_{]-\infty,x]} \,.\, P \in \mathcal{M}^{\searrow}(\nu)$, we are done.

Now the little Lemma 0.29, which we use several times, is immediate.

Proof of Lemma 0.29. Apply Remarks 0.48 and 0.35. □

Remark 0.49. Take $(H_i)_{i=1}^d$ increasing real functions, and $H = H_1 \otimes \cdots \otimes H_d : \mathbb{R}^d \to \mathbb{R}^d$. Then $H_\# : \mathrm{Marg}((\mu_i)_{i=1}^d) \to \mathrm{Marg}((H_i(\mu_i))_{i=1}^d)$ is contracting for ρ. To check it, think that $(F[H_\#P] - F[H_\#Q])(x_1, \ldots, x_d) = (H_\#P - H_\#Q)(\prod_i]-\infty, x_i])$ equals a term of the type $(H_\#P - H_\#Q)(\prod_i]-\infty, y_i\rceil)$, and use Proposition 0.43. Thus if $(P_n)_n$ converges to P in $\mathrm{Marg}(\mu_1, \ldots, \mu_d)$, $(H_\#P_n)_n$ converges to $H_\#P$ in $\mathrm{Marg}(H_1(\mu_1), \ldots, H_d(\mu_d))$.

The next remark is neither related to Proposition 0.43 nor to Lemma 0.29 but is an analogue of Remark 0.49 with $\preceq_{\mathrm{sto}}$ in place of ρ.

Remark 0.50. If P, Q are in $\mathcal{P}(\mathbb{R}^d)$, the $(H_i)_{i=1}^d$ are increasing functions, and $H \coloneqq H_1 \otimes \ldots \otimes H_d$ then $H_\# : \mathcal{P}(\mathbb{R}^d) \to \mathcal{P}(\mathbb{R}^d)$ is increasing for $\preceq_{\mathrm{lo}}$, i.e. $P \preceq_{\mathrm{lo}} Q \Rightarrow H_\#P \preceq_{\mathrm{lo}} H_\#Q$. To check it, think that

$$(F[H_\#P] - F[H_\#Q])(x_1, \ldots, x_d) = (H_\#P - H_\#Q)(\textstyle\prod_i]-\infty, x_i])$$

equals a term of the type $(H_\#P - H_\#Q)(\prod_i]-\infty, y_i\rceil)$, and, for the indices i such that "$\rceil$" stands for "[", use that $]-\infty, x_i[= \bigcup_n]-\infty, x_i - 1/n]$.

Remark 0.51. Notice then that $\rho(T, T') = \rho({}^{\mathrm{t}}T, {}^{\mathrm{t}}T')$ (see Definition 0.9 for ${}^{\mathrm{t}}T$). Thus, in Proposition 0.47, if $R \in \mathrm{Marg}(\mu', \nu)$ and if R has increasing kernel (i.e. by Remark 0.35, ${}^{\mathrm{t}}R$ maps $\mathcal{M}^{\searrow}(\nu)$ into $\mathcal{M}^{\searrow}(\mu')$):

$$\rho(PR, QR) = \rho({}^{\mathrm{t}}(PR), {}^{\mathrm{t}}(QR)) = \rho({}^{\mathrm{t}}R\,{}^{\mathrm{t}}P, {}^{\mathrm{t}}R\,{}^{\mathrm{t}}Q)) \leqslant \rho({}^{\mathrm{t}}P, {}^{\mathrm{t}}Q) = \rho(P, Q).$$

0.2.3 Proof of Lemma 0.28

First we prove Lemma 0.52, then its consequence Proposition 0.53, and finally Lemma 0.28.

Lemma 0.52. *For every $n \in \mathbb{N}$, let $P_n \in \mathrm{Marg}(\mu, \nu)$ have increasing kernel. Suppose moreover that $(P_n)_{n\in\mathbb{N}}$ converges to P_0. Let $h : [0,1] \to [0,1]$ be an increasing function and for every $n \in \mathbb{N}$, $\tilde{h}_n = P_n \,.\, h$. Then $(\tilde{h}_n)_{n\in\mathbb{N}}$ converges to $\tilde{h}_0$, μ-almost surely.*

Proof. By the equivalence proved in Proposition 3.11, the sequence $(\tilde{h}_n)_{n\in\mathbb{N}}$ is increasing. By Propositions 0.46(b) and 0.43, $\tilde{\rho}(P_n, P_0) \to_{n\to\infty} 0$, i.e. for every increasing g with values in $[0,1]$, $\int g(y)h(z)\mathrm{d}P_n(y,z) \to \int g(y)h(z)\mathrm{d}P_0(y,z)$. Hence:

$$\int g(x)\tilde{h}_n(x)\,\mathrm{d}\mu(x) \to \int g(x)\tilde{h}_0(x)\,\mathrm{d}\mu(x), \tag{14}$$

by definition of $\tilde{h}_n$. This also holds if g is the difference of two increasing functions, in particular $g = \mathbb{1}_{[a,b]} = \mathbb{1}_{[a,+\infty[} - \mathbb{1}_{[b,+\infty[}$.

Let A be the set of the elements x of $[0,1]$ such that $\mu[x,1]\mu[0,x] > 0$ and for every $n \in \mathbb{N}$, $\mu\text{-esssup}_{[0,x]}\tilde{h}_n = \mu\text{-essinf}_{[x,1]}\tilde{h}_n = \tilde{h}_n(x)$. In case $\mu = \lambda$, since $\tilde{h}_n$ is increasing, hence has at most a countable number of discontinuity points, $\mu(A)$ is 1. This also holds in the general case and is given by the increasing functions $\tilde{h}_n \circ G_\mu$, we leave the details to the reader. Now we take any $x \in A$ and prove $\tilde{h}_n(x) \to \tilde{h}_0(x)$. It is enough to prove $\limsup_n \tilde{h}_n(x) \leq \tilde{h}_0(x)$, since $\liminf_n \tilde{h}_n(x) \leq \tilde{h}_0(x)$ can be proved symmetrically. Suppose, for contradiction:

$$\limsup_n \tilde{h}_n(x) \geq \tilde{h}_0(x) + \varepsilon, \quad \text{for some } \varepsilon > 0.$$

As $\tilde{h}_0$ is increasing and $\tilde{h}_0(x) = \mu\text{-essinf}_{[x,1]}\tilde{h}_0$ there exists $y > x$ such that $\mu[x,y] > 0$ and:

$$\tilde{h}_0(x) \leq \frac{1}{\mu[x,y]} \int_{[x,y]} \tilde{h}_0 \,\mathrm{d}\mu \leq \tilde{h}_0(x) + \varepsilon/2.$$

This contradicts the facts that $\frac{1}{\mu[x,y]} \int_{[x,y]} \tilde{h}_n \,\mathrm{d}\mu \to_{n\to\infty} \frac{1}{\mu[x,y]} \int_{[x,y]} \tilde{h}_0 \,\mathrm{d}\mu$, obtained with $g = \mathbb{1}_{[x,y]}$ in (14), and that $\tilde{h}_n$ is increasing. □

Proposition 0.53. *Let P_n tend to P_0 in $\mathrm{Marg}(\mu_1, \ldots, \mu_d, \eta)$ and P'_n tend to P'_0 in $\mathrm{Marg}(\eta, \nu)$. Assume moreover that P'_n has increasing kernel for every n. Then $P'_n \circ P_n$ tends to $P'_0 \circ P_0$.*

Proof. By Propositions 0.46(b) and 0.43, we must show that

$$\int f(x)g(y)h(z)\mathrm{d}(P_n \circ P'_n)(x,y,z) \to \int f(x)g(y)h(z)\mathrm{d}(P_0 \circ P'_0)(x,y,z),$$

where $f((x_i)_{i=1}^d) = \prod_i f_i(x_i)$ and g, h and the f_i are any increasing functions from $\mathbb{R}$ to $[0,1]$. For all $n \geqslant 0$:

$$\begin{aligned}\int f(x)g(y)h(z)\mathrm{d}(P_n \circ P'_n)(x,y,z) &= \int f(x)g(y) \int h(z) k_{P'_n}(y, \mathrm{d}z)\mathrm{d}P_n(x,y) \\ &= \int f(x)(g\tilde{h}_n)(y)\mathrm{d}P_n,\end{aligned}$$

where $\tilde{h}_n = P_n \,.\, h$, by Definition 0.10 and Section 0.0.3.

By Lemma 0.52, $(g\tilde{h}_n)_n$ pointwise converges to $g\tilde{h}_0$, almost surely. Hence by the dominated convergence theorem:

$$\left| \int f(x)(g\tilde{h}_n)(y)\mathrm{d}P_0 - \int f(x)(g\tilde{h}_0)(y)\mathrm{d}P_0 \right| \to 0.$$

Moreover, $\tilde{\rho}(P_n, P_0)$ tends to zero and for every n, $g\tilde{h}_n$ is an increasing functions taking values in $[0,1]$. Hence, again by Propositions 0.46(b) and 0.43:

$$\left| \int f(x)(g\tilde{h}_n)(y)\mathrm{d}P_n - \int f(x)(g\tilde{h}_n)(y)\mathrm{d}P_0 \right| \to 0. \tag{15}$$

Therefore with the triangle inequality, (15) holds with $\tilde{h}_0$ in place of $\tilde{h}_n$ on the right, which is exactly what we claimed. □

We can now prove Lemma 0.28. It follows directly from Proposition 0.53.

Proof of Lemma 0.28. Consider a sequence $P^n_{1,2} \circ \cdots \circ P^n_{d,d+1}$ with limit $P \in \mathrm{Marg}(\mu_1, \ldots, \mu_{d+1})$ and denote $\mathrm{proj}^{i,i+1}_{\#} P$ by $P_{i,i+i}$. Since $(\mathrm{proj}^{i,i+1})_{\#}$ is continuous, $P_{i,i+1}$ is the limit of $(P^n_{i,i+1})_n$, hence $P_{i,i+1} \in \mathfrak{I}_{i,i+1}$. Proposition 0.53 used by induction shows that $P = P_{1,2} \circ \cdots \circ P_{n,n+1}$. □

0.3 Transitions kernels to and from the space of quantiles $[0,1]$

In this paper we will need to consider the quantile couplings of the measures μ_t with the reference measure $\lambda = \lambda|_{[0,1]}$.

Notation 0.54 (q_r, k_r, ${}^{\mathrm{t}}k_r$)**.** For all $r \in R$ we set $q_r = \mathfrak{Q}(\lambda, \mu_r)$; thus ${}^{\mathrm{t}}q_r$ (see Definition 0.9) is $\mathfrak{Q}(\mu_r, \lambda)$. Those couplings admit the respective disintegration kernels:

$$k_r : (\alpha, B) \longmapsto \delta_{G_r(\alpha)}(B),\ {}^{\mathrm{t}}k_r(x, \cdot) = \begin{cases} \delta_{F_r(x)} & \text{if } \mu_r(x) = 0 \\ (\mu_r(x))^{-1}\lambda\lfloor_{]F_r(x^-),F_r(x)[} & \text{if } \mu_r(x) > 0. \end{cases}$$

Remark 0.55. a. Notice that ${}^{\mathrm{t}}q_s \,.\, q_t = \mathrm{Joint}(\mu_s, {}^{\mathrm{t}}k_s \,.\, k_t) = \mathfrak{Q}^{s,t}$, where $\mathfrak{Q}^{s,t} = \mathfrak{Q}(\mu_s, \mu_t)$, see Definition 0.39. Indeed, ${}^{\mathrm{t}}q_s \,.\, q_t = {}^{\mathrm{t}}q_s \,.\, \mathrm{Id}_{\lambda,2} \,.\, q_t$, then apply (b) below, which will also be useful farther.

b. If some ordered pair (U, V) of variables has law $T \in \mathrm{Marg}(\lambda, \lambda)$, then

$$\mathrm{Marg}(\mu_s, \mu_t) \ni {}^{\mathrm{t}}q_s \,.\, T \,.\, q_t = \mathrm{Law}(G_s(U), G_t(V)) = (G_s \otimes G_t)_{\#}T.$$

Indeed, ${}^{\mathrm{t}}q_s.T.q_t = \mathrm{proj}^{1,4}_{\#}({}^{\mathrm{t}}q_s \circ T \circ q_t)$ and the 4-times process $(G_{\mu_s}(U), U, V, G_{\mu_t}(V))$ has law ${}^{\mathrm{t}}q_s \circ T \circ q_t$ and is Markov, since the σ-fields spanned by U and $\{U, G_{\mu_s}(U)\}$ are the same.

c. From (a) we get ${}^{\mathrm{t}}q_r.q_r = \mathrm{Id}_{\mu_r,2}$, so that ${}^{\mathrm{t}}q_r.q_r.{}^{\mathrm{t}}q_r = {}^{\mathrm{t}}q_r$ and $q_r.{}^{\mathrm{t}}q_r.q_r = q_r$. *However*, $q_r \,.\, {}^{\mathrm{t}}q_r \neq \mathrm{Id}_{\lambda,2}$. Indeed, $k_r \,.\, {}^{\mathrm{t}}k_r$ maps any quantile level $\alpha \in]0,1[$ on itself *except when* $G_r(\alpha)$ is an atom of μ_r. Actually, μ_r-almost surely:

$$k_r \,.\, {}^{\mathrm{t}}k_r(\alpha) = \begin{cases} \delta_\alpha & \text{if } \mu_r(G_r(\alpha)) = 0 \\ (\alpha^+ - \alpha^-)^{-1}\lambda|_{]\alpha^-,\alpha^+[} & \text{if } \alpha \in]\alpha^-, \alpha^+[, \end{cases}$$

where $]\alpha^-, \alpha^+[$ denotes any set $A_{r,x}$ as follows.

Notation 0.56 (Atomic levels, ℓ_r)**.** a. We denote by $A_{r,x}$ the interval $]F_r(x^-), F_r(x)[$ of quantile levels merged by G_r on some atom x of μ_r, and by A_r the set $\bigcup_{\mu_r(x)>0} A_{r,x} \subset [0,1]$ of "atomic levels" of μ_r.

b. We denote $k_r \,.\, {}^{\mathrm{t}}k_r$ described in Remark 0.55(c) by ℓ_r.

Remark 0.57. Measures $\theta \ll \lambda$ are transported by ℓ_r as follows: $\theta \,.\, \ell_r$ coincides with θ on $]0,1[\setminus A_r$, and on each $A_{r,x}$ it has constant density and mass $\theta(A_{r,x})$, i.e. equals $(\alpha^+ - \alpha^-)^{-1}\theta(A_{r,x})$. Equivalently, $F[\theta \,.\, \ell_r]$ is continuous, equal to $F[\theta]$ on $]0,1[\setminus A_{r,x}$, and affine on each connected component $A_{r,x}$ of A_r.

The product $\mathfrak{Q}^{s,r_1} \,.\, \mathfrak{Q}^{r_1,r_2} \,.\, \cdots \,.\, \mathfrak{Q}^{r_{m-1},r_m} \,.\, \mathfrak{Q}^{r_m,t}$ appearing in Theorem A is more deeply analyzed in Theorem A. Its kernel reads:

$$\begin{aligned} ({}^{\mathrm{t}}k_s \,.\, k_{r_1}) \,.\, ({}^{\mathrm{t}}k_{r_1} \,.\, k_{r_2}) \,.\, \cdots \,.\, ({}^{\mathrm{t}}k_{r_{m-1}} \,.\, k_{r_m}) \,.\, ({}^{\mathrm{t}}k_{r_m} \,.\, k_t) & \\ = {}^{\mathrm{t}}k_s \,.\, (k_{r_1} \,.\, {}^{\mathrm{t}}k_{r_1}) \,.\, \cdots \,.\, (k_{r_m} \,.\, {}^{\mathrm{t}}k_{r_m}) \,.\, k_t & \\ = {}^{\mathrm{t}}k_s \,.\, \ell_{r_1} \,.\, \cdots \,.\, \ell_{r_m} \,.\, k_t. & \end{aligned} \tag{16}$$

In Remark 0.66, (16) is further commented and re-expressed for transports in place of kernels. For now, it leads to introduce the following kernel.

Notation 0.58. Take $R = \{r_1, \dots, r_m\} \subset \mathbb{R}$. We denote the kernel $\ell_{r_1} \,.\, \ell_{r_2} \,.\, \cdots \,.\, \ell_{r_m}$ from $]0,1[$ to itself by ℓ_R, and $\mathrm{Joint}(\lambda; \ell_R) \in \mathrm{Marg}(\lambda, \lambda)$ by L_R. If $R = \varnothing$, $\ell_\varnothing$ is the identity kernel and $L_\varnothing$ the identity transport.

Notice that $\ell_{\{r\}} = \ell_r$ and that for any R, $\lambda \,.\, \ell_R = \lambda$. Moreover, ℓ_R only depends on $(A_r)_{r \in R}$. The following lemma is particularly simple.

Lemma 0.59. *Let $R \subset \mathbb{R}$ be a finite set and μ, ν be in $\mathcal{M}(\lambda)$.*

a. If $\mu \preceq_{\mathrm{sto}} \nu$ then $\mu \,.\, \ell_R \preceq_{\mathrm{sto}} \nu \,.\, \ell_R$, i.e. $\mathrm{Joint}(\lambda; \ell_R)$ has increasing kernel.

b. If μ is in $\mathcal{M}^{\searrow}(\lambda)$, then so is $\mu \,.\, \ell_R$, and $\mu \preceq_{\mathrm{sto}} \mu \,.\, \ell_R$.

Proof. For $r \in \mathbb{R}$, one easily checks that ℓ_r is increasing and stabilizes $\mathcal{M}^{\searrow}$; (a) and the first point of (b) follow. If $\mu \in \mathcal{M}^{\searrow}(\lambda)$, to see that $\mu \preceq_{\mathrm{sto}} \mu \,.\, \ell_R$, look at the cumulative distribution functions. They coincide off the components of A_r. Now on each of those, $F_{\mu.\ell_R}$ is affine whereas F_μ is concave since μ has decreasing density, so necessarily $F_\mu \geqslant F_{\mu.\ell_R}$. □

Though simple, the next lemma is a key of our construction of $\mathfrak{MQ}$.

Lemma 0.60. *Let $R \subset R'$ be two finite subsets of $\mathbb{R}$ and $\mu \in \mathcal{M}^{\searrow}(\lambda)$. Then $\mu \,.\, \ell_R \preceq_{\mathrm{sto}} \mu \,.\, \ell_{R'}$.*

Proof. Using an induction on the cardinal difference, it is enough to prove this if R' has one more element than R, say r'. We order it with the elements r_i of R: $r_1 < \cdots < r_k < r' < r_{k+1} < \cdots < r_m$. By Lemma 0.59(b), if μ is in $\mathcal{M}^{\searrow}(\lambda)$, so is $\mu_k \coloneqq \mu \,.\, \ell_{r_1} \,.\, \cdots \,.\, \ell_{r_k}$ and $\mu_k \preceq_{\mathrm{sto}} \mu_k \,.\, \ell_{r'}$. We apply $\ell_{r_{k+1}} \,.\, \cdots \,.\, \ell_{r_m}$ to each term of this inequality. Lemma 0.59(a) concludes. □

The choice of the suitable set $\mathcal{M}^{\searrow}$ in our monotonicity Lemma 0.60 (see also Remark 4.8) enables us to extend the definition of L_R to infinite sets R. Recall that $\preceq_{\mathrm{lo}}$ is interpreted in terms of $\preceq_{\mathrm{sto}}$ in Remark 0.38.

Lemma/Notation 0.61. *a. For all finite subsets R, R' of $\mathbb{R}$, $L_R \preceq_{\mathrm{lo}} \lambda \otimes \lambda$, and $R' \subset R \Rightarrow L_{R'} \preceq_{\mathrm{lo}} L_R$.*

b. For any $R \subset \mathbb{R}$, $\mathrm{losup}\{L_{R'} : R' \subset R \text{ and } R' \text{ finite}\}$ exists. We denote it by L_R (which is consistent with Notation 0.58 when R is finite).

c. For any $R \subset \mathbb{R}$, there is a nested sequence $(R_n)_n$ of finite subsets of R such that $(L_{R_n})_n$ converges weakly to L_R; if $(R_n)_n$ satisfies this property, all sequence $(R'_n)_n$ such that $R'_n \supset R_n$ also does.

Proof. Let us prove (a). For all $(x, y) \in [0,1]^2$, $F[L_R](x, y) = (\lambda\lfloor_{[0,x]}).L_R([0, y])$. After Lemma 0.59(b), $(\lambda\lfloor_{[0,x]}) \,.\, L_R \in \mathcal{M}^{\searrow}$ so $f : y \mapsto \frac{1}{y}((\lambda\lfloor_{[0,x]}) \,.\, L_R)([0, y])$ is decreasing, thus $f(y) \geq f(1) = x$. Hence:

$$\forall\, y \in [0,1], \quad ((\lambda\lfloor_{[0,x]}) \,.\, L_R)([0, y]) \geqslant xy = F[\lambda \otimes \lambda](x, y), \quad \text{i.e. } L_R \preceq_{\mathrm{lo}} \lambda \otimes \lambda.$$

Now if $R' \subset R$ for every x we can apply Lemma 0.60 to $\lambda\lfloor_{[0,x]} \in \mathcal{M}^{\searrow}$. This yields $F[L_R](x, y) = (\lambda\lfloor_{[0,x]}) \,.\, L_R([0, y]) \geqslant (\lambda\lfloor_{[0,x]}) \,.\, L_{R'}([0, y]) = F[L_{R'}](x, y)$, which is the expected relation for $\preceq_{\mathrm{lo}}$. Then, (b) and (c) follow from criterion (b) of Lemma 0.32. Indeed, by (a), $\mathcal{S} = \{L_{R'} : R' \subset R \text{ and } R' \text{ finite}\}$ is bounded from above by $\lambda \otimes \lambda$, and if $\{L_{R'_1}, L_{R'_2}\} \subset \mathcal{S}$, then $L_{R'_1 \cup R'_2} \in \mathcal{S}$ and $L_{R'_1 \cup R'_2} \succeq_{\mathrm{lo}} L_{R'_i}$ for $i = 1, 2$. Finally, the assertion about $(R'_n)_n$ follows from (a) and the interpretation of both ρ and $\preceq_{\mathrm{sto}}$ with cumulative distribution functions. □

Notation 0.62. For all $R \subset \mathbb{R}$, we denote by ℓ_R the kernel associated with L_R. This is again consistent with Notation 0.56(b) when R is finite.

The following result is crucial to define processes on $(]0,1[,\lambda)$ with Corollary 0.15, as is done in particular in Definition 0.26 for Lev_R and Lev.

Proposition 0.63. *If R and R' are subsets of $\mathbb{R}$ such that $r \leqslant r'$ for all $(r,r') \in R \times R'$, then : $L_{R\cup R'} = L_R \,.\, L_{R'}$. In particular, for $s < t < u$, $L_{[s,u]} = L_{[s,t]} \,.\, L_{[t,u]}$.*

Proof. Using Lemma 0.61(b) and (c) we find sequences $(R_n)_n$ and $(R'_n)_n$ of finite subsets of R and R' respectively, such that L_{R_n} converges weakly to L_R, $L_{R'_n}$ to $L_{R'}$, and $L_{R_n\cup R'_n}$ to $L_{R\cup R'}$. Besides, since $r \leqslant r'$ for all $(r,r') \in R\times R'$, since the R_n are finite and since $L_{\{r\}}$ is idempotent (so that in case $R\cap R' \neq \varnothing$ and $R_n \cap R'_n = R\cap R' = \{r\}$, a repetition of $L_{\{r\}}$ does not matter), $L_{R_n\cup R'_n} = L_{R_n} \,.\, L_{R'_n}$. Then, using Proposition 0.43 and the distance ρ introduced in it:

$$\begin{aligned}\rho(L_R \,.\, L_{R'}, L_{R\cup R'}) &\leqslant \rho(L_R \,.\, L_{R'}, L_R \,.\, L_{R'_n}) \\ &\qquad + \rho(L_R \,.\, L_{R'_n}, L_{R_n} \,.\, L_{R'_n}) + \rho(L_{R_n\cup R'_n}, L_{R\cup R'}) \\ &\leqslant \rho(L_{R'}, L_{R'_n}) + \rho(L_R, L_{R_n}) + \rho(L_{R_n\cup R'_n}, L_{R\cup R'})\end{aligned}$$

by Proposition 0.47 and Remark 0.51, since ${}^{\mathrm{t}}L_{R'_n}$ also preserves $\mathcal{M}^{\searrow}(\lambda)$. All terms tend to zero when n tends to infinity. The desired equality follows. □

0.4 The Markov-quantile process $\mathfrak{MQ}$; proof of Theorem A

Reminder 0.64. For $(s,t) \in \mathbb{R}^2$ with $t > s$, and $R = \{r_1, \dots, r_m\} \subset [s,t]$, $\mathfrak{Q}^{s,t}_{[R]}$ denotes the coupling $\mathfrak{Q}(\mu_s,\mu_{r_1}).\mathfrak{Q}(\mu_{r_2},\mu_{r_3}).\cdots.\mathfrak{Q}(\mu_{r_m},\mu_t) \in \mathrm{Marg}(\mu_s,\mu_t)$, see Theorem A(A(a)(iv)).

Proposition/Definition 0.65.

a. *The set $\{\mathfrak{Q}^{s,t}_{[R]} \mid R \subset \mathbb{R}$ and $\sharp R < \infty\}$ has a lower orthant supremum and there is a nested sequence $(R_n)_{n\in\mathbb{N}}$ of finite sets such that $\mathfrak{Q}^{s,t}_{[R_n]}$ tends to it. We denote it by $\mu_{s,t}$.*

b. *The family $(\mu_{s,t})_{s<t}$ is consistent in the sense of Definition 0.16, giving rise to a Markov measure $\mathfrak{MQ} \in \mathrm{Marg}((\mu_t)_{t\in\mathbb{R}})$, that we call the* Markov-quantile measure *attached to $(\mu_t)_{t\in\mathbb{R}}$.*

c. *For all s and $t > s$, $\mathfrak{MQ}^{s,t} = (G_s \otimes G_t)_{\#}L_{]s,t[} = (G_s \otimes G_t)_{\#}L_{[s,t]}$.*

To show Proposition 0.65 we first state the following crucial relation between $\mathfrak{Q}^{s,t}_{[R]} \in \mathrm{Marg}(\mu_s,\mu_t)$ and $L_R \in \mathrm{Marg}(\lambda,\lambda)$ defined in Section 0.3.

Remark 0.66. Take any finite subset R of $\mathbb{R}$ and (s,t) with $s < t$, then:

$$\mathfrak{Q}^{s,t}_{[R]} = (G_s \otimes G_t)_{\#}L_{]s,t[\cap R} = (G_s \otimes G_t)_{\#}L_{[s,t]\cap R}.$$

Indeed, $\mathfrak{Q}^{s,t} = {}^{\mathrm{t}}q_s \,.\, q_t = \mathrm{Law}(G_s, G_t)$, see Remark 0.55(a). It also equals $(G_s \otimes G_t)_{\#}(\mathrm{Id}_2)$, where $\mathrm{Id}_2 = \mathrm{Id}_{\lambda,2}$ is the identity transport form λ to itself, see

Notation 0.8, since by Remark 0.55(b), $\mathfrak{Q}(\mu_s,\mu_t) = {}^{\mathrm{t}}q_s \,.\, q_t = {}^{\mathrm{t}}q_s.\mathrm{Id}_2 \,.\, q_t = (G_s \otimes G_t)_{\#}(\mathrm{Id}_2)$. More generally, for any $\{r_1,\dots,r_m\} \subset]s,t[$:

$$\begin{aligned}\mathfrak{Q}(\mu_s,\mu_{r_1}) \,.\, \cdots \,.\, \mathfrak{Q}(\mu_{r_m},\mu_t) &= {}^{\mathrm{t}}q_s \,.\, q_{r_1} \,.\, {}^{\mathrm{t}}q_{r_1} \,.\, q_{r_2} \,.\, \cdots \,.\, {}^{\mathrm{t}}q_{r_{m-1}} \,.\, q_{r_m} \,.\, {}^{\mathrm{t}}q_{r_m} \,.\, q_t \\ &= (G_s \otimes G_t)_{\#}(q_{r_1} \,.\, {}^{\mathrm{t}}q_{r_1} \,.\, q_{r_2} \,.\, \cdots \,.\, {}^{\mathrm{t}}q_{r_{m-1}} \,.\, q_{r_m} \,.\, {}^{\mathrm{t}}q_{r_m}) \\ &\qquad\qquad \text{(notice that this writing involves neither } {}^{\mathrm{t}}q_s \text{ nor } q_t) \\ &= (G_s \otimes G_t)_{\#} L_{\{r_1,\dots,r_m\}}.\end{aligned}$$

Besides recall that $q_s \,.\, {}^{\mathrm{t}}q_s \,.\, q_s = q_s$ and ${}^{\mathrm{t}}q_t \,.\, q_t \,.\, {}^{\mathrm{t}}q_t = {}^{\mathrm{t}}q_t$, so that:

$$\begin{aligned}\mathfrak{Q}(\mu_s,\mu_{r_1}) \,.\, \cdots \,.\, \mathfrak{Q}(\mu_{r_m},\mu_t) &= (G_s \otimes G_t)_{\#}(q_s \,.\, {}^{\mathrm{t}}q_s \,.\, q_{r_1} \,.\, \cdots \,.\, {}^{\mathrm{t}}q_{r_m} \,.\, q_t \,.\, {}^{\mathrm{t}}q_t) \\ &= (G_s \otimes G_t)_{\#} L_{\{s,r_1,\dots,r_m,t\}}.\end{aligned}$$

Proof of Proposition 0.65. We have only to gather our results. We set

$$\mathcal{S} \coloneqq \{R \subset]s,t[\,:\, R \text{ is finite}\}.$$

By Remark 0.50 and Lemma 0.61(a),

$$\begin{aligned}&\forall\, R \in \mathcal{S}, \quad (G_s \otimes G_t)_{\#} L_R \preceq_{\mathrm{lo}} (G_s \otimes G_t)_{\#}(\lambda \otimes \lambda), \\ &R' \subset R \implies (G_s \otimes G_t)_{\#} L_{R'} \preceq_{\mathrm{lo}} (G_s \otimes G_t)_{\#} L_R. \qquad \text{and}\end{aligned}$$

Thus, by Lemma 0.32(b), $\operatorname{losup}_{R' \in \mathcal{S}} (G_s \otimes G_t)_{\#} L_{R'}$ exists and is the limit of some sequence $((G_s \otimes G_t)_{\#} L_{R_n})_n$. Hence the limit in (a) is given by Remark 0.66:

$$\mathfrak{Q}^{s,r_1} \,.\, \mathfrak{Q}^{r_1,r_2} \,.\, \cdots \,.\, \mathfrak{Q}^{r_{m(n)-1},r_{m(n)}} \,.\, \mathfrak{Q}^{r_{m(n)},t} = (G_s \otimes G_t)_{\#} L_{R_n}.$$

We prove now the first part of (c), i.e. $\mu_{s,t} = (G_s \otimes G_t)_{\#} L_{]s,t[}$ and, at the same time, that the sequence $(R_n)_n$ in (a) can be chosen to be nested. Let $(R_n)_n$ be a nested sequence of $\mathcal{S}$ such that $L_{R_n} \to L_{]s,t[}$, i.e. $F[L_{R_n}]$ pointwise converges to $F[L_{]s,t[}]$. If $M \in \operatorname{Marg}(\mu_s,\mu_t)$ satisfies $(G_s \otimes G_t)_{\#} L_{R'} \preceq_{\mathrm{lo}} M$ for every $T \in \mathcal{S}$, this also holds for $R' = R_n$ for all n. Now by Remark 0.49, $(G_s \otimes G_t)_{\#} L_{R_n} \to (G_s \otimes G_t)_{\#} L_{]s,t[}$. Therefore, going to the limits at the level of the cumulative distribution functions we get $(G_s \otimes G_t)_{\#} L_{]s,t[} \preceq_{\mathrm{lo}} M$. Then Remark 0.50 gives

$$(G_s \otimes G_t)_{\#} L_{R'} \preceq_{\mathrm{lo}} (G_s \otimes G_t)_{\#} L_{]s,t[},$$

hence:

$$(G_s \otimes G_t)_{\#} L_{]s,t[} = \operatorname{losup}_{R' \subset \mathbb{R} \text{ and } R' \text{ finite}} \big\{ \mathfrak{Q}^{s,t}_{[R']} = (G_s \otimes G_t)_{\#} L_{R'} \big\}.$$

Corollary 0.15 gives (b). Indeed, Proposition 0.63 on the composition of transports L_R gives the consistency of $(\mu_{s,t})_{s,t}$ (see Definition 0.16):

$$(G_s \otimes G_t)_{\#} L_{]s,t[} = {}^{\mathrm{t}}q_s \,.\, L_{]s,t[} \,.\, q_t \quad \text{and} \quad (G_t \otimes G_u)_{\#} L_{]t,u[} = {}^{\mathrm{t}}q_t L_{]t,u[} q_u.$$

Since $q_t \,.\, {}^{\mathrm{t}}q_t = L_{\{t\}}$, $\mu_{s,u} = \mu_{s,t} \,.\, \mu_{t,u}$.

For the second equality of (c), proceed as at the end of Remark 0.66. $\square$

Theorem 0.67. *Let $(\mu_t)_{t\in\mathbb{R}}$ be a family of probability measures on $\mathbb{R}$. In points* (b) *and* (4.21(c)), *$(R_n)_n$ stands for a nested sequence of finite subsets of $\mathbb{R}$ and R for $\bigcup_n R_n$.*

a. There is a countable set $R\subset\mathbb{R}$ satisfying:

$$\text{for all } (s,t) \text{ with } s<t,\ L_{R\cap]s,t[}=L_{]s,t[}. \tag{17}$$

b. If R satisfies (17) *then (see Reminder 0.2 for the weak convergence):*

$$(\mathfrak{Q}_{[R_n]})_n \text{ converges weakly to } \mathfrak{MQ}. \tag{18}$$

Lemma 0.68. *Let $\mathcal{T}$ denote a totally ordered set of indices (in practice, $\mathcal{T}=\mathbb{R}^+$ or $\mathcal{T}=\mathbb{N}$). If $(R_\tau)_{\tau\in\mathcal{T}}$ is a family of subsets of $\mathbb{R}$, increasing for the inclusion, setting $R=\bigcup_\tau R_\tau$, $(L_{R_\tau})_{\tau\in\mathcal{T}}$ is increasing for $\preceq_{\rm lo}$ and tends weakly to L_R when τ tends to infinity.*

Proof. It rests on this remark: by definition of $\preceq_{\rm lo}$ in Definition 0.21 and of ρ in Proposition 0.43, if A, A', A'' are measures of $\mathcal{P}(\mathbb{R}^d)$ with the same marginals and $A\preceq_{\rm lo}A'\preceq_{\rm lo}A''$, then $\rho(A',A'')\leqslant\rho(A,A'')$. Now we prove the lemma. By definition of L_{R_τ} in Lemma 0.61(b) the sequence $(L_{R_\tau})_{\tau\in\mathcal{T}}$ is increasing, be the sets R_τ finite or not. By Lemma 0.61(c), and as ρ metrizes the weak topology (Proposition 0.43), for all $\varepsilon>0$ we find a finite $R'\subset R$ such that $\rho(L_R,L_{R'})\leqslant\varepsilon$. As $R=\bigcup_\tau R_\tau$ there is a τ_0 such that $R_{\tau_0}\supset R'$. Then: $\tau\geqslant\tau_0\Rightarrow L_{R'}\preceq_{\rm lo}L_{R_\tau}\preceq_{\rm lo}L_R\Rightarrow\rho(L_{R'},L_{R_\tau})\leqslant\varepsilon$ by the remark. □

Lemma 0.69. *There is a nested sequence $(R_n)_{n\in\mathbb{N}}$ of finite subsets of $\mathbb{R}$ such that $s<t\Rightarrow L_{R_n\cap]s,t[}\to L_{]s,t[}$*

Proof. Let $((u_k,u'_k))_{k\in\mathbb{N}}$ be a dense sequence in

$$\{(x,y)\in\mathbb{R}^2:\ x<y\}$$

and for all $n\geq1$, $(R^n_k)_{k\in\mathbb{N}}$ a nested sequence of finite subsets of $\mathbb{R}$ such that $\rho(L_{R^n_k\cap]u_k,u'_k[},L_{]u_k,u'_k[})\leqslant1/n$ for every $k\in\mathbb{N}$. Denote $\bigcup_{k=1}^n R^n_k$ by R_n. Then for any s and $t>s$, $L_{R_n\cap]s,t[}\to_{n\to\infty}L_{]s,t[}$. Let us prove it. Fix $\varepsilon>0$. By Lemma 0.68 (use a subsequence $(k_n)_n$ such that $(]u_{k_n},u'_{k_n}[)_n$ is an exhaustion of $]s,t[$), there exists k such that $\rho(L_{]s,t[},L_{]u_k,u'_k[})<\varepsilon/2$. Then $L_{]u_k,u'_k[\cap R_n}\preceq_{\rm lo}L'\preceq_{\rm lo}L_{]s,t[}$, where L' stands for $L_{]s,t[\cap R_n}$ or $L_{]u_k,u'_k[}$ for any k. For $n\geq\max\{k,2/\varepsilon\}$:

$$\rho(L_{]s,t[},L_{]u_k,u'_k[\cap R_n})\leq\rho(L_{]s,t[},L_{]u_k,u'_k[})+\rho(L_{]u_k,u'_k[},L_{]u_k,u'_k[\cap R_n})<\varepsilon.$$

Therefore with $L'=L_{]s,t[\cap R_n}$ we obtain the desired convergence to $L_{]s,t[}$. □

Definition 0.70. Let I be an interval. If, for some interval $J\supset I$ such that $J\setminus I$ is disconnected (then for all such smaller J'), $L_J\neq L_{J\setminus I}$, we call I an *essential atomic interval of* $(\mu_t)_{t\in\mathbb{R}}$. If $I=\{t\}$ is essential, we call t an *essential atomic time of* $(\mu_t)_{t\in\mathbb{R}}$.

To check the parenthesis in the definition, suppose that $L_{J'}=L_{J'\setminus I}$ and get $L_J=L_{J\setminus I}$ by Proposition 0.63.

Proposition 0.71. *If a nested sequence $(R_n)_n$ is as in Lemma 0.69 then $\bigcup_n R_n$ contains all the essential atomic times of $(\mu_t)_t$. In particular, these times are at most countably many.*

Proof. We show a contrapositive result. Suppose that t is an essential atomic time, that $s < t$ and $s' > t$ are such that $L_{]s,s'[\setminus\{t\}} \neq L_{]s,s'[}$, and that $(R_n)_n$ is a nested sequence of finite sets such that $t \notin \bigcup_n R_n$, that $L_{R_n\cap]s,t[} \to L_{]s,t[}$ and that $L_{R_n\cap]t,s'[} \to L_{]t,s'[}$. Then:

$$\begin{aligned} L_{R_n\cap]s,s'[} &= L_{R_n\cap]s,t[} \,.\, L_{R_n\cap]s,t[} \quad \text{by Proposition 0.63} \\ &\to L_{]s,t[} \,.\, L_{]t,s'[} \quad \text{by assumption and by Proposition 0.53} \\ &= L_{]s,s'[\setminus\{t\}} \neq L_{]s,s'[} \quad \text{by Proposition 0.63 and the assumption.} \end{aligned}$$

Therefore, $(R_n)_n$ cannot be as in Lemma 0.69. □

We can now prove Theorem 0.67(a).

Proof of Theorem 0.67(a). Take $(R_n)_n$ a sequence as given by Lemma 0.69. Then, for any s and $t > s$:

$$\begin{aligned} L_{R\cap]s,t[} &= \operatorname{losup}\{L_{R_n\cap]s,t[}\} \quad \text{by Lemma 0.68} \\ &\succeq_{\text{lo}} \operatorname{losup}\{L_{R'_n\cap]s,t[}\} \quad \text{as } R'_n \subset R_n \text{ and by Lemma 0.61(a)} \\ &= \lim_{n\to\infty} L_{R'_n\cap]s,t[} = L_{]s,t[} \quad \begin{array}{l}\text{by Lemma 0.68 and} \\ \text{by property of the sets } R'_n.\end{array} \end{aligned}$$

But by definition, $L_{R\cap]s,t[} \preceq_{\text{lo}} L_{]s,t[}$, thus $L_{R\cap]s,t[} = L_{]s,t[}$. □

To provide a clear proof of Theorem 0.67(b) we introduce an auxiliary notion in Definition 0.72 below. Suppose that some $P \in \operatorname{Marg}(\mu, \mu_1, \dots, \mu_k)$ is disintegrated as $P = \operatorname{Joint}(\mu, k_P)$ and that $g : \mathbb{R} \to \mathbb{R}$ and $h : \mathbb{R}^k \to \mathbb{R}^k$ are measurable maps. May we disintegrate $(g\otimes h)_\# P$? In case g is into one easily checks $(g \otimes \operatorname{Id})_\# P = \operatorname{Joint}(g_\#\mu, k_P^g)$, where k_P^g is defined by $k_P^g(y, \cdot) = k_P^g(g^{-1}(y), \cdot)$, $g_\#\mu$-almost surely. Otherwise, the next notion and lemma will enable us to obtain a similar disintegration, and associated properties.

Definition 0.72. We say that $g : \mathbb{R} \to \mathbb{R}$ *fits* $P \in \operatorname{Marg}(\mu, \mu_1, \dots, \mu_k)$ if there exists a kernel k_P^g such that $k_P^g(g(x), \cdot) = k_P(x, \cdot)$, μ-almost surely.

Remark 0.73.

a. If g fits $P = \operatorname{Joint}(\mu, k)$ and h is a measurable map from $\mathbb{R}^k$ into itself we can disintegrate $(g \otimes h)_\# P$ as $\operatorname{Joint}(g_\#\mu, h_\# k_P^g)$, where $h_\# k_P^g(y, \cdot) = k_P^g(y, h^{-1}(\cdot))$.

b. If g fits P, it also fits $P \,.\, P'$ and $P \circ P'$.

c. If g fits $P \in \operatorname{Marg}(\mu, \nu_1, \dots, \nu_k)$ or $Q \in \operatorname{Marg}(\mu, \nu'_1, \dots, \nu'_{k'})$, and if $f : \mathbb{R}^k \to \mathbb{R}^k$ and $h : \mathbb{R}^{k'} \to \mathbb{R}^{k'}$ are measurable maps, then:

$$(f \otimes g \otimes h)_\#({}^{\text{t}}P \circ Q) = (f \otimes g)_\# {}^{\text{t}}P \circ (g \otimes h)_\# Q.$$

The proofs are direct, using the definitions of a kernel, composition and concatenation in Section 0.0.4. Notice that, in case $\mu = \lambda\lfloor_{[0,1]}$ and g is a quantile function (which is the only case in which we will use the remark), point (c) is a particular case of Lemma 0.75(b) below. In the language of this lemma, "g fits ${}^{\text{t}}P$ or Q" means that ${}^{\text{t}}P \,.\, L = {}^{\text{t}}P$ or $L \,.\, Q = Q$, which both imply in particular that ${}^{\text{t}}P \,.\, L \,.\, Q = {}^{\text{t}}P \,.\, Q$.

We will need the following little technical result.

Lemma 0.74. *Take $a < b$ in $\mathbb{R}$ and f and g two positive bounded increasing functions from $]a,b[$ to $\mathbb{R}$. Then*

$$\frac{1}{b-a}\int_a^b fg\,\mathrm{d}\lambda \geq \left(\frac{1}{b-a}\int_a^b f\mathrm{d}\lambda\right)\left(\frac{1}{b-a}\int_a^b g\mathrm{d}\lambda\right),$$

with equality if and only if f or g is constant.

Proof. The measures $\mu = f\lambda$ and $\nu = \frac{1}{b-a}\int_a^b f\,\mathrm{d}\lambda$ have the same mass $\int_a^b f\,\mathrm{d}\lambda$. Besides, $\nu \preceq_{\mathrm{sto}} \mu$; indeed, for any $c \in]a,b[$, since f is increasing:

$$\frac{1}{c-a}\mu(]-\infty,c]) = \frac{1}{c-a}\int_a^c f\,\mathrm{d}\lambda \leq \frac{1}{b-a}\int_a^b f\,\mathrm{d}\lambda = \frac{1}{c-a}\nu(]-\infty,c]).$$

Hence, as g is increasing, $\int g\,\mathrm{d}\mu \geq \int g\,\mathrm{d}\nu$, equivalent to the wished inequality.

Plainly, equality occurs if f or g is constant. If f is not constant, we in fact proved that equality in $\int g_c\mathrm{d}\mu \leq \int g_c\mathrm{d}\nu$ holds for $g_c = \mathbb{1}_{[c,b]}$ if and only if $c \in \{a,b\}$. This remains true for positive combinations of functions g_c and, being a little careful, for limits of them. □

Lemma 0.75. *Take $\mu \in \mathcal{P}(\mathbb{R})$, denote by $g = G_\mu$ its quantile function and $F = F_\mu$ its cumulative distribution function. Take $L = \mathfrak{Q}(\lambda,\mu).\mathfrak{Q}(\mu,\lambda)$ and, similarly as in Notation 0.56, let $(b_i)_{i\in\mathcal{I}}$ be the atoms of μ and for each i, A_i be the interval $]F(b_i^-), F(b_i)[$ of quantile levels merged by g on b_i. Finally set $A = \bigcup_{i\in\mathcal{I}} A_i$. Moreover, take*

$$((\nu_i)_i, (\nu'_i)_i) \in \mathcal{P}(\mathbb{R})^{k+k'},$$
$$P \in \mathrm{Marg}(\lambda\lfloor_{[0,1]}, \nu_1 \otimes \ldots \otimes \nu_k) \quad \textit{and} \quad Q \in \mathrm{Marg}(\lambda_{[0,1]}, \nu'_1 \otimes \ldots \otimes \nu'_{k'}).$$

Suppose that P and Q have increasing kernel (for $\preceq_{\mathrm{lo}}$ in place of $\preceq_{\mathrm{sto}}$ if $k \geq 2$ or $k' \geq 2$). Then:

a. ${}^{\mathrm{t}}P \,.\, L \,.\, Q$ equals ${}^{\mathrm{t}}P \,.\, Q$ if and only if for each $i \in \mathcal{I}$, at least one of the two kernel functions $x \mapsto k_P(x,\ \cdot\ )$ or $x \mapsto k_Q(x,\ \cdot\ )$ is constant on A_i,

b. For any measurable maps $f : \mathbb{R}^k \to \mathbb{R}^k$ and $h : \mathbb{R}^{k'} \to \mathbb{R}^{k'}$,

$${}^{\mathrm{t}}P \,.\, L \,.\, Q = {}^{\mathrm{t}}P \,.\, Q \implies (f \otimes g \otimes h)_\#({}^{\mathrm{t}}P \circ Q) = (f \otimes g)_\#{}^{\mathrm{t}}P \circ (g \otimes h)_\# Q.$$

Proof. We recall particularly here that we have throughout in mind the analogy between composition of transport plans, or of their kernels, and product of matrices, hinted at in Remark 0.5. In the following, $B_1 \subset \mathbb{R}^k$ and $B_3 \subset \mathbb{R}^{k'}$ stand for any sets of the type $\prod_j]-\infty, d_j]$ and B_2 for any interval $]-\infty, b]$; (a,b,c) stand for variables in the target of (f,g,h), and (x,y,z) for variables in their sources. The coupling L, equal to ${}^{\mathrm{t}}q_r \,.\, q_r$ (with $\mu = \mu_r$) introduced in Remark 0.55, is described in this remark. It is such that ${}^{\mathrm{t}}P \,.\, L \,.\, Q = ({}^{\mathrm{t}}P \,.\, L) \,.\, (L \,.\, Q) = {}^{\mathrm{t}}((g \otimes \mathrm{Id}_k)_\# P) \,.\, ((g \otimes \mathrm{Id}_{k'})_\# Q)$. In turn, for any measurable functions

$f : \mathbb{R}^k \to \mathbb{R}^k$ and $h : \mathbb{R}^{k'} \to \mathbb{R}^{k'}$:

$$\begin{aligned}
&\int_{B_2} k_{(g\otimes f)_\# P}(b, B_1) \,.\, k_{(g\otimes h)_\# Q}(b, B_3)\,\mathrm{d}\mu(b) \\
&\quad= \sum_{i\in\mathcal{I},\, b_i\in B_2} \mu(b_i)[k_{(g\otimes f)_\# P}(b_i, B_1) \,.\, k_{(g\otimes h)_\# Q}(b_i, B_3)] \\
&\qquad\quad + \int_{B_2\setminus\bigcup_i\{b_i\}} k_P(g^{-1}(b), f^{-1}(B_1)) \,.\, k_Q(g^{-1}(b), h^{-1}(B_3))\,\mathrm{d}\mu(b) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{as } g \text{ is injective outside of } A \\
&\quad= \sum_{i\in\mathcal{I},\, b_i\in B_2} \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, f^{-1}(B_1))\,\mathrm{d}\lambda(y) \int_{A_i} k_Q(y, h^{-1}(B_3))\,\mathrm{d}\lambda(y) \\
&\qquad\quad + \int_{g^{-1}(B_2)\setminus A} k_P(y, f^{-1}(B_1)) \,.\, k_Q(y, h^{-1}(B_3))\,\mathrm{d}\lambda(y).
\end{aligned} \tag{19}$$

Then, using for instance the expression (2) given in Definition 0.10 of the concatenation, and the fact that for any transport plans R and R', $R \,.\, R' = \mathrm{proj}^{1,3}_\#(R \circ R')$ we get:

$$\begin{aligned}
&({}^{\mathrm{t}}P \,.\, Q - {}^{\mathrm{t}}((g \otimes \mathrm{Id})_\# P) \,.\, (g \otimes \mathrm{Id})_\# Q)(B_1 \times B_3) \\
&\quad= \int_{[0,1]} k_P(y, B_1) k_Q(y, B_3)\mathrm{d}y - \int_{\mathbb{R}} k_{(g\otimes\mathrm{Id})_\# P}(b, B_1) k_{(g\otimes\mathrm{Id})_\# Q}(b, B_3)\mathrm{d}\mu(b) \\
&\quad= \sum_{i\in\mathcal{I}} \int_{A_i} k_P(y, B_1) k_Q(y, B_3)\mathrm{d}y - \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, B_1)\mathrm{d}y \int_{A_i} k_Q(y, B_3)\mathrm{d}y
\end{aligned}$$

by (19) with $B_2 = \mathbb{R}$, $f = \mathrm{Id}$, $h = \mathrm{Id}$—notice in particular that the two terms of the difference, viewed as integrals on $[0,1]$, differ possibly only on $A \subset [0,1]$. Now, after Lemma 0.74 applied on each A_i with $f_i : y \mapsto k_P(y, B_1)$ and $g_i : y \mapsto k_Q(y, B_3)$, which are increasing since P and Q have increasing kernel, each term of the sum is non-negative. Thus, equality holds if and only if each vanishes, which again by Lemma 0.74 means that for each i, f_i or g_i is constant. This, holding for any B_1 and B_3, means point (a). For point (b):

$$\begin{aligned}
&{}^{\mathrm{t}}((g \otimes f)_\# P) \circ ((g \otimes h)_\# Q)(B_1 \times B_2 \times B_3) \\
&\qquad= \int_{B_2} k_{(f\otimes g)_\# P}(b, B_1) k_{(g\otimes h)_\# Q}(b, B_3)\,\mathrm{d}\mu(b) \quad \text{by (2)} \\
&\qquad= \int_{g^{-1}(B_2)\setminus A} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y \\
&\qquad\quad + \sum_{i\in\mathcal{I},\, b_i\in B_2} \frac{1}{\lambda(A_i)} \int_{A_i} k_P(y, f^{-1}(B_1))\mathrm{d}y \int_{A_i} k_Q(y, h^{-1}(B_3))\mathrm{d}y \quad \text{by (19)},
\end{aligned}$$

but on each A_i, by (a), $k_P(\,.\,, f^{-1}(B_1))$ or $k_Q(\,.\,, h^{-1}(B_3))$ is a constant function, so the last term may be written:

$$\sum_{i\in\mathcal{I},\, b_i\in B_2} \int_{A_i} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y,$$

and therefore the initial expression equals $\int_{g^{-1}(B_2)} k_P(y, f^{-1}(B_1)) k_Q(y, h^{-1}(B_3))\mathrm{d}y$, that is to say $(f \otimes g \otimes h)_\#({}^{\mathrm{t}}P \circ Q)(B_1 \times B_2 \times B_3)$. □

Let us prove point (b). Take $R \subset \mathbb{R}$ and Lev_R and Lev in $\mathrm{Marg}((\lambda_t)_{t\in\mathbb{R}})$ given by Definition 0.26. We need the following claim, that extends Remark 0.27, and relies eventually on Remark 0.55(b):

Claim. *If R is finite, $G_\#\mathrm{Lev}_R = \mathfrak{Q}_{[R]}$, where $G = (\otimes_{t\in\mathbb{R}} G_t)$.*

Let us prove it. Take $R \subset \mathbb{R}$ finite and n its cardinal. We must prove that for any finite $S = \bigcup_{i=1}^N \{s_i\}$, $G_\#\mathrm{Lev}_R^{s_1,\dots,s_N} = \mathfrak{Q}_{[R]}^{s_1,\dots,s_N}$, where, by an abuse of notation we will often make use of, G stands for $\otimes_{i=1}^N G_{s_i}$. It suffices to prove it in the case $S \supset R$, which we suppose now. We introduce the cardinals $\ell_0, \dots, \ell_n$ of the subsets of $S \setminus R$ situated between to consecutive elements r_k and r_{k+1} of $R \cup \{\pm\infty\}$. We re-index these subsets as $\{s_1^k, \dots, s_{\ell_k}^k\}$. Since Lev_R is Markov, $(\mathrm{proj}^S)_\#\mathrm{Lev}_R = B_0 \circ \cdots \circ B_m$, where:

$$B_k = L_{r_k} \circ \mathrm{Id}_{\ell_k} \circ L_{r_{k+1}} = \begin{cases} \mathrm{Id}_{\ell_0} \circ L_{r_1} & \text{if } k = 0 \\ L_{r_m} \circ \mathrm{Id}_{\ell_m} & \text{if } k = m \\ L_{[r_k, r_{k+1}]} = L_{r_k}.L_{r_{k+1}} & \text{if } \ell_k = 0 \\ L_{[r_k, r_{k+1}]} = L_{r_k} \circ L_{r_{k+1}} & \text{if } \ell_k = 1 \\ L_{r_k} \circ \underbrace{\mathrm{Id}_2 \circ \cdots \circ \mathrm{Id}_2}_{\ell_k - 1} \circ L_{r_{k+1}} & \text{otherwise,} \\ & \text{i.e. if } \ell_k \geq 2, \end{cases}$$

hence we must prove that: $G_\#(B_1 \circ \cdots \circ B_m) = \mathfrak{Q}_{[R]}^{s_1,\dots,s_N}$. For all $r \in R$, the quantile function $G_r = G_{\mu_r}$ fits $q_r = \mathrm{Joint}(\mu_r, x \mapsto \delta_{G_r(x)})$, so we may take $k_{q_r}^{G_r} : x \mapsto \delta_x$ as given by Definition 0.72. Now, by Remark 0.73(b), it also fits $L_{\{r\}} = {}^{\mathrm{t}}L_{\{r\}} = q_r \,.\, {}^{\mathrm{t}}q_r$. Therefore, by Remark 0.73(b) and (c), $G_\#(B_1 \circ \cdots \circ B_m) = B_1' \circ \ldots \circ B_m'$, where:

$$B_k' = G_\# B_k = \begin{cases} (\otimes_{s \leqslant r_1} G_s)_\# \mathrm{Id}_{\ell_0} \circ L_{r_1} & \text{if } k = 0 \\ (\otimes_{s \geqslant r_m} G_s)_\# L_{r_m} \circ \mathrm{Id}_{\ell_m} & \text{if } k = m \\ (G_{r_k} \otimes G_{r_{k+1}})_\# (L_{r_k} \,.\, L_{r_{k+1}}) & \text{if } \ell_k = 0 \\ (G_{r_k} \otimes G_{s_1^k} \otimes G_{r_{k+1}})_\# (L_{r_k} \circ L_{r_{k+1}}) & \text{if } \ell_k = 1 \\ (\otimes_{r_k \leqslant s \leqslant r_{k+1}} G_s)_\# (L_{r_k} \circ \mathrm{Id}_{\ell_k} \circ L_{r_{k+1}}) & \text{otherwise, i.e. } \ell_k \geq 2. \end{cases}$$

Proving that $B_k' = \mathfrak{Q}(\mu_{r_k}, \mu_{s_1^k}, \dots, \mu_{s_{\ell_k - 1}^k}, \mu_{r_{k+1}})$ will now prove the claim. For simplicity we assume $k \notin \{0, m\}$ and $\ell_k \geq 1$ but the other cases, which are simpler, can be proved similarly. Observe that:

- by definition of $\mathfrak{Q}$ and G,

$$\mathfrak{Q}(\mu_{r_k}, \mu_{s_1^k}, \dots, \mu_{s_{\ell_k}^k}, \mu_{r_{k+1}}) = \mathrm{Law}((G_{r_k}(U), \dots, G_{r_{k+1}}(U)),$$

  where U is a variable of law λ on $[0,1]$,

- $(\otimes_{r_k \leqslant s \leqslant r_{k+1}} G_s)_\# (L_{r_k} \circ \mathrm{Id}_{\ell_k} \circ L_{r_{k+1}})$ is the law of

$$(G_{r_k}(U_1), G_{s_1^k}(U_2), \dots, G_{s_{\ell_k}^k}(U_2), G_{r_{k+1}}(U_3)),$$

  where $\mathrm{Law}(U_1, U_2, \dots, U_2, U_3) = B_k$. In particular, $\mathrm{Law}(U_i) = \lambda$ for $i = 1, 2, 3$, $\mathrm{Law}(U_1, U_2) = L_{r_k}$, and $\mathrm{Law}(U_2, U_3) = L_{r_{k+1}}$.

So only the first and last variables may differ. Now, notice that by Remark 0.55(b) applied to $T = L_{\{r\}} = q_r.{}^{\mathrm{t}}q_r$, if two variables (U, V) satisfy $\mathrm{Law}(U, V) = L_{\{r\}}$, the law of $(G_r(U), G_r(V))$ is

$$ {}^{\mathrm{t}}q_r \,.\, L_{\{r\}} \,.\, q_r = {}^{\mathrm{t}}q_r \,.\, q_r \,.\, {}^{\mathrm{t}}q_r \,.\, q_r = {}^{\mathrm{t}}q_r \,.\, q_r = \mathrm{Id}_{\mu_r,2}, $$

therefore $G_r(U) = G_r(V)$ almost surely. Using this with $r = r_k$ and $(U, V) = (U_1, U_2)$, respectively $r = r_{k+1}$ and $(U, V) = (U_2, U_3)$, we get respectively $G_{r_k}(U_1) = G_{r_k}(U_2)$ and $G_{r_{k+1}}(U_2) = G_{r_{k+1}}(U_3)$ almost surely. The claim is proved.

Now take $R \subset \mathbb{R}$ satisfying (17) and $(R_n)_n$ a nested exhaustion of it by finite sets. Let $\widetilde{\mathrm{Lev}} \in \mathrm{Marg}((\lambda_t)_{t\in\mathbb{R}})$ be the Markov process having $\widetilde{\mathrm{Lev}}^{s,t} = L_{R\cap[s,t]}$ as 2-marginals. Recall that $L_{R\cap[s,t]} = L_{R\cap\{s\}} \,.\, L_{R\cap]s,t[} \,.\, L_{R\cap\{t\}} = L_{R\cap\{s\}} \,.\, L_{]s,t[} \,.\, L_{R\cap\{t\}} = \lim_{n\to\infty} \mathrm{Lev}_{R_n\cap[s,t]} = \lim_{n\to\infty} \mathrm{Lev}^{s,t}_{R_n}$. It exists by Corollary 0.15, whose assumption is satisfied by Proposition 0.63. All these transport plans have increasing kernel, hence by Lemma 0.28, for every $S = \bigcup_{i=1}^k \{s_i\}$:

$$ \mathrm{Lev}^{s_1,\dots,s_k}_{R_n} = \mathrm{Lev}^{s_1,s_2}_{R_n} \circ \cdots \circ \mathrm{Lev}^{s_{k-1},s_k}_{R_n} \to \widetilde{\mathrm{Lev}}^{s_1,s_2} \circ \cdots \circ \widetilde{\mathrm{Lev}}^{s_{k-1},s_k} = \widetilde{\mathrm{Lev}}^{s_1,\dots,s_k}. \quad (20) $$

Thus Lev_{R_n} converges weakly to $\widetilde{\mathrm{Lev}}$, and then by Remark 0.49, $G_\#\mathrm{Lev}_{R_n} \to G_\#\widetilde{\mathrm{Lev}}$. We are left with the tasks to prove $G_\#\mathrm{Lev}_{R_n} = \mathfrak{Q}_{[R_n]}$ and $G_\#\widetilde{\mathrm{Lev}} = \mathfrak{MQ}$. The former is our claim above. Let us prove the latter.

Note. At the beginning of Section 1 we announced (21), i.e. $G_\#\mathrm{Lev} = \mathfrak{MQ}$. In fact, we prove $G_\#\widetilde{\mathrm{Lev}} = \mathfrak{MQ}$, which is a bit more difficult. To get (21) the same arguments work, the final reasoning with Lemma 0.75(b) being replaced by a direct use of Remark 0.73(c), as for all $\{r_1, \dots, r_\ell\}$, G_{r_ℓ} fits $\mathrm{Lev}^{r_1,\dots,r_\ell}$.

We recall that $\mathfrak{MQ}$ was defined as the unique Markov law with the same marginals of dimension 2 as $G_\#\mathrm{Lev}$. But by (17) and Proposition 0.65(c), the 2-marginals of $G_\#\widetilde{\mathrm{Lev}}$ and $G_\#\mathrm{Lev}$ are equal. Hence it is sufficient to prove that $G_\#\widetilde{\mathrm{Lev}}$ is Markov, i.e. that for all $(s_1, \dots, s_k)$, $(G_\#\widetilde{\mathrm{Lev}})^{s_1,\dots,s_k} = (G_\#\widetilde{\mathrm{Lev}})^{s_1,s_2} \circ \dots \circ (G_\#\widetilde{\mathrm{Lev}})^{s_{1-1},s_k}$. Since $\widetilde{\mathrm{Lev}}$ is Markov, $(\widetilde{\mathrm{Lev}})^{s_2,\dots,s_k} = \widetilde{\mathrm{Lev}}^{s_2,s_3} \circ \dots \circ \widetilde{\mathrm{Lev}}^{s_{k-1},s_k}$; besides, notice the following fact, that we will prove a bit below:

Fact. For all $(s_i)_{i=1}^k \in \mathbb{R}^k$, $\mathrm{Lev}^{s_1,\dots,s_k}$ viewed as a transport plan from $(\mathbb{R}, \lambda)$ to $(\mathbb{R}^{k-1}, \lambda^{\otimes k-1})$ has increasing kernel (for the order $\preceq_{\mathrm{lo}}$ instead of $\preceq_{\mathrm{sto}}$).

Hence we may conclude by using $k-1$ times Lemma 0.75(b). Let us check the first step. Since the measures ${}^{\mathrm{t}}(\widetilde{\mathrm{Lev}}^{s_1,s_2})$ and $\widetilde{\mathrm{Lev}}^{s_2,\dots,s_k}$ have increasing kernel, we have only to show that $\widetilde{\mathrm{Lev}}^{s_1,s_2}.\widetilde{\mathrm{Lev}}^{s_2,s_3} = \widetilde{\mathrm{Lev}}^{s_1,s_2} \,.\, L_{\{s_2\}}.\widetilde{\mathrm{Lev}}^{s_2,s_3}$, i.e. by definition of $\widetilde{\mathrm{Lev}}$, that $L_{R\cap[s_1,s_2]} \,.\, L_{R\cap[s_2,s_3]} = L_{R\cap[s_1,s_2]} \,.\, L_{s_2} \,.\, L_{R\cap[s_2,s_3]}$. This amounts to checking that:

$$ \begin{cases} L_{\rfloor s_1,s_2[} \,.\, L_{]s_2,s_3\lceil} = L_{\rfloor s_1,s_2[} \,.\, L_{s_2} \,.\, L_{]s_2,s_3\lceil} & \text{if } s_2 \notin R \\ L_{\rfloor s_1,s_2]} \,.\, L_{[s_2,s_3\lceil} = L_{\rfloor s_1,s_2]} \,.\, L_{s_2} \,.\, L_{[s_2,s_3\lceil} & \text{otherwise.} \end{cases} $$

The second point is true by Proposition 0.63, and if $s_2 \notin R$, by Proposition 0.71, s_2 is not an essential atomic time, which is the wanted equality. We finally

must prove the fact stated above. Actually it is true for any concatenation $P_1 \circ \ldots \circ P_k$ of couplings P_i from $\mathbb{R}$ to $\mathbb{R}$ with increasing kernel —and $\mathrm{Lev}^{s_1,\ldots,s_k}$ is of this type, see (20). We check this for $k = 2$; the same argument, applied by induction, gives the general case. Take B_2 and B_3 two intervals of the type $]-\infty, a]$ and $x \leqslant x'$; we must show that: $k_{P_1 \circ P_2}(x, B_2 \times B_3) \geqslant k_{P_1 \circ P_2}(x', B_2 \times B_3)$. As P_2 has increasing kernel, the function $k_{P_2}(\,\cdot\,, B_3)$ is decreasing (thus also $\mathbb{1}_{B_2} k_{P_2}(\,\cdot\,, B_3)$). As P_1 has increasing kernel, $k_{P_1}(x, \,\cdot\,) \preceq_{\mathrm{sto}} k_{P_1}(x', \,\cdot\,)$. Thus:

$$k_{P_1 \circ P_2}(x, B_2 \times B_3) = \int k_{P_1}(x, \mathrm{d}y) \mathbb{1}_{B_2} k_{P_2}(y, B_3) \geqslant \int k_{P_1}(x', \mathrm{d}y) \mathbb{1}_{B_2} k_{P_2}(y, B_3),$$

the desired result. This proves part (b) of the theorem.

1 Construction and characterization of the Markov-quantile process

Now $(\mu_t)_{t\in\mathbb{R}}$ is a family of probability measures on $\mathbb{R}$, F_t denotes the cumulative distribution function F_{μ_t} of μ_t and G_t its quantile function, see Section 0.1. In this section we build the Markov-quantile measure $\mathfrak{MQ}$ and prove Theorems A and B. Our proof of Theorems A–A is based on transport plans $L_{[s,t]} \in \mathrm{Marg}(\lambda, \lambda)$, defined in Section 0.3, that are the 2-marginals of an important auxiliary process law called the quantile level measure $\mathrm{Lev} \in \mathrm{Marg}(\lambda_t)_{t\in\mathbb{R}}$, where each λ_t is a copy indexed by t of $\lambda = \lambda_{[0,1]}$. Here is the link between Lev and $\mathfrak{MQ}$: G_t maps $([0,1], \lambda_t)$ to $(\mathbb{R} \cup \{\pm\infty\}, \mu_t)$; set $G = \otimes_{t\in\mathbb{R}} G_t$, so that $G_\#$ maps $\mathrm{Marg}((\lambda_t)_{t\in\mathbb{R}})$ to $\mathrm{Marg}((\mu_t)_{t\in\mathbb{R}})$. Then, as proved in the proof Theorem 0.67 p. 27:

$$G_\# \mathrm{Lev} = \mathfrak{MQ}. \tag{21}$$

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